

ENGINEERING MATHEMATICS - I

UNIT - 2 : DIFFERENTIAL CALCULAS - I

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By Gulshan sir

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UNIT-2 : Differential Calculus - I

➤ Successive Differentiation (n^{th} order derivatives)

➤ Leibnitz theorem

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➤ Partial derivatives

➤ Euler's Theorem for Homogeneous functions

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➤ Total derivative

➤ Curve tracing

Gateway Classes

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 1

Today's target

- Successive Differentiation (Part-1)
- PYQs
- DPP



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Successive Differentiation

The process of finding differentiation of a function again and again is called successive differentiation

$$y = f(x)$$

$$\frac{dy}{dx} = f'(x)$$

$$\frac{d^2y}{dx^2} = f''(x)$$

$$\frac{d^3y}{dx^3} = f'''(x)$$

$$\vdots$$

$$\frac{d^ny}{dx^n} = f^n(x)$$

$$\frac{d^{n+1}y}{dx^{n+1}} = f^{n+1}(x)$$

$$\frac{d^{n+2}y}{dx^{n+2}} = f^{n+2}(x)$$

$$D = \frac{d}{dx}$$

Function	First derivative	Second Derivative	Third Derivative	n^{th} Derivative	$(n+1)^{\text{th}}$ Derivative	$(n+2)^{\text{th}}$ Derivative
$y = f(x)$	$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$	$\frac{d^ny}{dx^n}$	$\frac{d^{n+1}y}{dx^{n+1}}$	$\frac{d^{n+2}y}{dx^{n+2}}$
	y_1	y_2	y_3	y_n	y_{n+1}	y_{n+2}
	Dy	D^2y	D^3y	D^ny	$D^{n+1}y$	$D^{n+2}y$
	$f'(x)$	$f''(x)$	$f'''(x)$	$f^n(x)$	$f^{n+1}(x)$	$f^{n+2}(x)$

v.v imp
v.v. imp

Q. 1 if $y = \cos^{-1} x$, prove that $(1 - x^2)y_2 - xy_1 = 0$

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$$y = \cos^{-1} x$$

diff. wrt x

$$y_1 = \frac{-1}{\sqrt{1-x^2}}$$

$$y_1 \sqrt{1-x^2} = -1$$

Squaring both side

$$y_1^2 (1-x^2) = 1$$

Again diff. wrt x

$$y_1^2 (-2x) + (1-x^2) 2y_1 y_2 = 0$$

$$2y_1 [-xy_1 + (1-x^2)y_2] = 0$$

$$-xy_1 + (1-x^2)y_2 = 0$$

$$(1-x^2)y_2 - xy_1 = 0$$

Hence proved

1. n^{th} Derivative of $y = x^m$, where $n < m$

$$y = x^m$$

$$y_1 = m x^{m-1}$$

$$y_2 = m(m-1) x^{m-2}$$

$$y_3 = m(m-1)(m-2) x^{m-3}$$

$$\vdots$$

$$y_n = m(m-1)(m-2) \dots [m-(n-1)] x^{m-n}$$

If $m = n$

$$y_n = n(n-1)(n-2) \dots [\cancel{n-n+1}] x^{n-n}$$

$$y_n = n(n-1)(n-2) \dots \underline{3 \cdot 2 \cdot 1}$$

$$y_n = n!$$

If $n > m$

$$y_n = 0$$

2. n^{th} derivative of $y = (ax + b)^m$, where $n < m$

$$y = (ax + b)^m$$

$$y_1 = m(ax + b)^{m-1} a$$

$$y_2 = m(m-1)(ax + b)^{m-2} a^2$$

$$y_3 = m(m-1)(m-2)(ax + b)^{m-3} a^3$$

$$\vdots$$

$$y_n = m(m-1)(m-2) \dots (m-n+1) (ax + b)^{m-n} a^n$$

If $m = n$

$$y_n = n(n-1)(n-2) \dots (n-n+1) (ax + b)^0 a^n$$

$$y_n = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 a^n$$

$$y_n = a^n \times n! \quad \checkmark$$

If $n > m$

$$y_n = 0$$

3. n^{th} derivative of $y = a^{mx}$

$$y = a^{mx}$$

$$y_1 = m a^{mx} \log a$$

$$y_2 = m^2 a^{mx} (\log a)^2$$

$$y_3 = m^3 a^{mx} (\log a)^3$$

$\vdots$

$$y_n = m^n a^{mx} (\log a)^n$$

If $m = 1$

$$y_n = a^x (\log a)^n$$

4. n^{th} derivative of $y = e^{mx}$

$$y = e^x$$

$$y_n = m^n e^{mx} (\log e)^n$$

$$y_n = m^n e^{mx}$$

If $m = 1$

$$y_n = e^x$$

5. n^{th} derivative of $y = \frac{1}{ax+b}$

$$y = (ax+b)^{-1}$$

$$y_1 = -1(ax+b)^{-2} a$$

$$y_2 = (-1)(-2)(ax+b)^{-3} a^2$$

$$y_3 = (-1)(-2)(-3)(ax+b)^{-4} a^3$$

⋮

$$y_n = (-1)(-2)(-3) \dots (-n)(ax+b)^{-(n+1)} a^n$$

$$y_n = \frac{(-1)^n 1 \times 2 \times 3 \dots n \times a^n}{(ax+b)^{n+1}}$$

$$y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

If $a=1, b=0$

$$y = \frac{1}{x}$$

$$y_n = \frac{(-1)^n n!}{x^{n+1}}$$

Partial Fraction : For any rational function $\frac{f(x)}{g(x)}$

- if degree $f(x) < \text{degree } g(x)$, then partial fraction occur
- if degree $f(x) \geq \text{degree } g(x)$, then first divide $f(x)$ by $g(x)$

Type - 1 :
$$\frac{1}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

Type - 2 :
$$\frac{1}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{(x-b)}$$

Type - 3 :
$$\frac{1}{(ax^2+bx+c)(x-d)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{x-d}$$

Q.2 If $y = \frac{1}{1-5x+6x^2}$, find y_n

$$y = \frac{1}{1-5x+6x^2}$$

$$y = \frac{1}{6x^2-5x+1}$$

$$y = \frac{1}{6x^2-3x-2x+1}$$

$$y = \frac{1}{3x(2x-1)-1(2x-1)}$$

$$y = \frac{1}{(2x-1)(3x-1)}$$

$$y = \frac{1}{(2x-1)(3 \times \frac{1}{2} - 1)} + \frac{1}{(3x-1)(2 \times \frac{1}{3} - 1)}$$

$$y = \frac{2}{2x-1} - \frac{3}{3x-1}$$

$$y_n = 2 \left[\frac{d^n}{dx^n} \left(\frac{1}{2x-1} \right) \right] - 3 \left[\frac{d^n}{dx^n} \left(\frac{1}{3x-1} \right) \right]$$

$$\frac{d^n}{dx^n} \left(\frac{1}{ax+b} \right) = \frac{(-1)^n a^n n!}{(ax+b)^{n+1}}$$

$$y_n = 2 \left[\frac{(-1)^n 2^n n!}{(2n-1)^{n+1}} \right] - 3 \left[\frac{(-1)^n 3^n n!}{(3n-1)^{n+1}} \right]$$

$$y_n = \frac{(-1)^n 2^{n+1} n!}{(2n-1)^{n+1}} - \frac{(-1)^n 3^{n+1} n!}{(3n-1)^{n+1}}$$

Q.3 Find the n^{th} derivative of $\frac{2x+1}{(2x-1)(2x+3)}$

$$y = \frac{2n+1}{(2n-1)(2n+3)}$$

$$y = \frac{\cancel{2}x(\frac{1}{\cancel{2}}) + 1}{(2n-1)(\cancel{2}x\frac{1}{\cancel{2}} + 3)} + \frac{\cancel{2}x(-\frac{3}{\cancel{2}}) + 1}{(2n+3)(\cancel{2}x(-\frac{3}{\cancel{2}}) - 1)}$$

$$y = \frac{2}{4(2n-1)} + \frac{(-2)}{(-4)(2n+3)}$$

$$y = \frac{1}{2} \left[\frac{1}{2n-1} + \frac{1}{2n+3} \right]$$

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$$y_n = \frac{1}{2} \left[\frac{(-1)^n n! 2^n}{(2n-1)^{n+1}} + \frac{(-1)^n n! 2^n}{(2n+3)^{n+1}} \right]$$

$$y_n = \frac{(-1)^n n! 2^n}{2} \left[\frac{1}{(2n-1)^{n+1}} + \frac{1}{(2n+3)^{n+1}} \right]$$

$$y_n = (-1)^n n! 2^{n-1} \left[\frac{1}{(2n-1)^{n+1}} + \frac{1}{(2n+3)^{n+1}} \right]$$

Q.4 Find the n^{th} derivative of

$$\frac{1}{x^2+a^2}$$

OR

If $y = \frac{1}{x^2+a^2}$, prove that $y_n = \frac{(-1)^n n! \sin(n+1)\theta \sin^{n+1}\theta}{a^{n+2}}$

$$y = \frac{1}{x^2+a^2}$$

$$y = \frac{1}{(n+ia)(n-ia)}$$

$$y = \frac{1}{(n+ia)(-2ia)} + \frac{1}{(n-ia)(2ia)}$$

$$y = \frac{1}{2ia} \left[\frac{1}{n-ia} - \frac{1}{n+ia} \right]$$

$$a^2+b^2 = (a+ib)(a-ib)$$

$$y_n = \frac{1}{2ia} \left[\frac{(-1)^n \times n!}{(n-ia)^{n+1}} - \frac{(-1)^n \times n!}{(n+ia)^{n+1}} \right]$$

$$y_n = \frac{(-1)^n \times n!}{2ia} \left[\frac{1}{(n-ia)^{n+1}} - \frac{1}{(n+ia)^{n+1}} \right] \quad \checkmark$$

Put $x = r \cos \theta$

$a = r \sin \theta$

$\theta = \tan^{-1}\left(\frac{a}{x}\right)$

$$y_n = \frac{(-1)^n \times n!}{2ia} \left[\frac{1}{(r \cos \theta - i r \sin \theta)^{n+1}} - \frac{1}{(r \cos \theta + i r \sin \theta)^{n+1}} \right]$$

$$y_n = \frac{(-1)^n \times n!}{2ia \times r^{n+1}} \left[\frac{1}{(\cos \theta - i \sin \theta)^{n+1}} - \frac{1}{(\cos \theta + i \sin \theta)^{n+1}} \right]$$

$$y_n = \frac{(-1)^n \times n!}{2ia \times r^{n+1}} \left[(\cos \theta - i \sin \theta)^{-(n+1)} - (\cos \theta + i \sin \theta)^{-(n+1)} \right] \checkmark$$

using De Moivre's theorem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$y_n = \frac{(-1)^n \times n!}{2ia r^{n+1}} \left[\cancel{\cos(n+1)\theta} + i \sin(n+1)\theta - \cancel{\cos(n+1)\theta} + i \sin(n+1)\theta \right]$$

$$y_n = \frac{(-1)^n \times n!}{2ia r^{n+1}} \times \cancel{2} i \sin(n+1)\theta$$

$$y_n = \frac{(-1)^n \times n!}{a \times \left(\frac{a}{\sin\theta}\right)^{n+1}} \times \sin(n+1)\theta$$

$$y_n = \frac{(-1)^n \times n! \times \sin(n+1)\theta \sin^{n+1}\theta}{a^{n+2}}$$

Q.5 Find the n^{th} derivative of $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$

OR

$$\frac{d}{dn} \left(\tan^{-1} n \right) = \frac{1}{1+n^2}$$

If $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$, show that $y_n = 2 (-1)^{n-1} (n-1)! \sin n\theta \sin^n \theta$

$$y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$$

Put $x = \tan \theta$

$$y = \tan^{-1} \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)$$

$$y = \tan^{-1} (\tan 2\theta)$$

$$y = 2\theta$$

$$y = 2 \tan^{-1} x$$

$$y_1 = \frac{2}{1+x^2}$$

$$y_1 = \frac{2}{(x+i)(x-i)}$$

$$y_1 = \frac{2}{(x+i)(-i)} + \frac{2}{(x-i)(i)}$$

$$y_1 = \frac{1}{i(x-i)} - \frac{1}{i(x+i)}$$

$$y_1 = \frac{1}{i} \left[\frac{1}{x-i} - \frac{1}{x+i} \right]$$

$$y_n = \frac{1}{i} \left[\frac{(-1)^{n-1} (n-1)!}{(x-i)^n} - \frac{(-1)^{n-1} (n-1)!}{(x+i)^n} \right]$$

$$y_n = \frac{(-1)^{n-1}(n-1)!}{i} \left[\frac{1}{(x-i)^n} - \frac{1}{(x+i)^n} \right]$$

Put

$$x = r \cos \theta$$

$$1 = r \sin \theta$$

$$\theta = \tan^{-1}\left(\frac{1}{r}\right)$$

$$y_n = \frac{(-1)^{n-1}(n-1)!}{i} \left[\frac{1}{(r \cos \theta - i r \sin \theta)^n} - \frac{1}{(r \cos \theta + i r \sin \theta)^n} \right]$$

$$y_n = \frac{(-1)^{n-1}(n-1)!}{i r^n} \left[\frac{1}{(\cos \theta - i \sin \theta)^n} - \frac{1}{(\cos \theta + i \sin \theta)^n} \right]$$

$$y_n = \frac{(-1)^{n-1}(n-1)!}{i r^n} \left[(\cos \theta - i \sin \theta)^{-n} - (\cos \theta + i \sin \theta)^{-n} \right]$$

Using De Moivre's Theorem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{i^n} \left[\cancel{\cos n\theta} + i \sin n\theta - \cancel{\cos n\theta} + i \sin n\theta \right]$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{i^n} \times 2i \sin n\theta$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{\left(\frac{1}{\sin\theta}\right)^n} \times 2 \sin n\theta$$

$$y_n = 2 (-1)^{n-1} (n-1)! \sin n\theta \sin^n \theta$$



DPP - 1

Topic : Successive Differentiation (Part-1)

Q.1 if $y = \sin(asin^{-1}x)$, prove that $(1 - x^2)y_2 - xy_1 = -a^2y$

Q.2 If $y = \frac{5x+12}{6+5x+x^2}$, find y_n

Q.3 Find the n^{th} derivative of $\frac{x^2}{(x+2)(2x+3)}$

Q.4 Find the n^{th} derivative of $\tan^{-1}\left(\frac{1+x}{1-x}\right)$

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 2

Today's target

- Successive Differentiation (Part-2)
- PYQs
- DPP



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(6) n^{th} derivative of $y = \log(ax + b)$

$$y = \log(ax + b)$$

$$y_1 = \frac{1}{(ax + b)} \times a$$

$$y_1 = (ax + b)^{-1} a$$

$$y_2 = (-1)(ax + b)^{-2} a^2$$

$$y_3 = (-1)(-2)(ax + b)^{-3} a^3$$

⋮

$$y_n = (-1)(-2) \dots \dots \dots [-(n-1)] (ax + b)^{-n} a^n$$

$$y_n = (-1)^{n-1} \underbrace{1 \times 2 \dots \dots (n-1)}_{(n-1)!} (ax + b)^{-n} a^n$$

$$y_n = \frac{(-1)^{n-1} (n-1)! a^n}{(ax + b)^n} \quad \checkmark \checkmark$$

If $a = 1, b = 0$

$$y = \log x$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{x^n}$$

Q.1 Find the n^{th} derivative of $\log(ax + x^2)$

$$y = \log(ax + x^2)$$

$$y = \log x(x+a)$$

$$y = \log x + \log(x+a)$$

$$y_n = \frac{d^n}{dx^n}(\log x) + \frac{d^n}{dx^n}\{\log(x+a)\}$$

$$y_n = \frac{(-1)^{n-1}(n-1)!}{x^n} + \frac{(-1)^{n-1}(n-1)!}{(x+a)^n}$$

$$\textcircled{1} \log mn = \log m + \log n$$

$$\textcircled{2} \log \frac{m}{n} = \log m - \log n$$

$$\textcircled{3} \log m^n = n \log m$$

$$y_n = (-1)^{n-1}(n-1)! \left[\frac{1}{x^n} + \frac{1}{(x+a)^n} \right]$$

(7) n^{th} derivative of $y = \sin(ax + b)$

$$y = \sin(ax + b)$$

$$y_1 = a \cos(ax + b) = a \sin(ax + b + \frac{\pi}{2})$$

$$y_2 = a^2 \cos(ax + b + \frac{\pi}{2}) = a^2 \sin(ax + b + \frac{2\pi}{2})$$

$$y_3 = a^3 \cos(ax + b + \frac{2\pi}{2}) = a^3 \sin(ax + b + \frac{3\pi}{2})$$

⋮

⋮

$$y_n = a^n \sin(ax + b + \frac{n\pi}{2})$$

$$\text{If } b = 0$$

$$y = \sin ax$$

$$y_n = a^n \sin(ax + \frac{n\pi}{2})$$

$$\text{If } a = 1, b = 0$$

$$y = \sin x$$

$$y_n = \sin(x + \frac{n\pi}{2})$$

(8) n^{th} derivative of $y = \cos(ax + b)$

$$y = \cos(ax + b)$$

$$y_1 = -a \sin(ax + b) = a \cos(ax + b + \frac{\pi}{2})$$

$$y_2 = -a^2 \sin(ax + b + \frac{\pi}{2}) = a^2 \cos(ax + b + \frac{2\pi}{2})$$

$$y_3 = -a^3 \sin(ax + b + \frac{2\pi}{2}) = a^3 \cos(ax + b + \frac{3\pi}{2})$$

$$y_n = a^n \cos(ax + b + \frac{n\pi}{2})$$

If $b = 0$

$$y = \cos ax$$

$$y_n = a^n \cos(ax + \frac{n\pi}{2})$$

If $a = 1, b = 0$

$$y = \cos x$$

$$y_n = \cos(x + \frac{n\pi}{2})$$

Q.2 Find the n^{th} derivative of $y = \sin^3 x$

$$y = \sin^3 x$$

$$\therefore \sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\sin^3 x = \frac{3 \sin x - \sin 3x}{4}$$

$$y = \frac{1}{4} (3 \sin x - \sin 3x)$$

$$y_n = \frac{1}{4} \left[3 \frac{d^n}{dx^n} (\sin x) - \frac{d^n}{dx^n} (\sin 3x) \right]$$

$$y_n = \frac{1}{4} \left[3 \sin \left(x + \frac{n\pi}{2} \right) - 3^n \sin \left(3x + \frac{n\pi}{2} \right) \right]$$

$$y_n = \frac{3}{4} \sin \left(x + \frac{n\pi}{2} \right) - \frac{3^n}{4} \sin \left(3x + \frac{n\pi}{2} \right)$$

Q.3 Find the n^{th} derivative of $y = \sin x \sin 2x$

$$y = \sin x \sin 2x$$

$$y = \frac{1}{2} (2 \sin x \sin 2x)$$

$$\therefore 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$y = \frac{1}{2} [\cos(x-2x) - \cos(x+2x)]$$

$$y = \frac{1}{2} [\cos x - \cos 3x]$$

$$y_n = \frac{1}{2} \left[\frac{d^n}{dx^n} (\cos x) - \frac{d^n}{dx^n} (\cos 3x) \right]$$

$$y_n = \frac{1}{2} \left[\cos\left(x + \frac{n\pi}{2}\right) - 3^n \cos\left(3x + \frac{n\pi}{2}\right) \right]$$

Q.4 If $\mu = \sin nx + \cos nx$ prove that $\mu_r = n^r [1 + (-1)^r \sin 2nx]^{\frac{1}{2}}$ where μ_r denotes the r^{th} differential co-efficient of μ w r t x . Hence show that $\mu_8(\pi) = \left(\frac{1}{2}\right)^{\frac{31}{2}}$ when $n = \frac{1}{4}$ **AKTU 2018**

$$\mu = \sin nx + \cos nx$$

$$\mu_r = \frac{d^r}{dx^r}(\sin nx) + \frac{d^r}{dx^r}(\cos nx)$$

$$\mu_r = n^r \sin\left(nx + \frac{r\pi}{2}\right) + n^r \cos\left(nx + \frac{r\pi}{2}\right)$$

$$\mu_r = n^r \left[\sin\left(nx + \frac{r\pi}{2}\right) + \cos\left(nx + \frac{r\pi}{2}\right) \right]$$

$$\mu_r = n^r \left[\left\{ \sin\left(nx + \frac{r\pi}{2}\right) + \cos\left(nx + \frac{r\pi}{2}\right) \right\}^2 \right]^{\frac{1}{2}}$$

$$\mu_r = n^r \left[\sin^2\left(nx + \frac{r\pi}{2}\right) + \cos^2\left(nx + \frac{r\pi}{2}\right) + 2 \sin\left(nx + \frac{r\pi}{2}\right) \cos\left(nx + \frac{r\pi}{2}\right) \right]^{\frac{1}{2}}$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$2 \sin \theta \cos \theta = \sin 2\theta$$

$$u_x = n^x \left[1 + \sin 2 \left(nx + \frac{x\pi}{2} \right) \right]^{\frac{1}{2}}$$

$$u_x = n^x \left[1 + \sin(2nx + x\pi) \right]^{\frac{1}{2}}$$

$$\therefore \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$u_x = n^x \left[1 + \sin 2nx \cos x\pi + \cos 2nx \sin x\pi \right]$$

$$\therefore \cos n\pi = (-1)^n$$

$$\therefore \sin n\pi = 0$$

$$u_x = n^x \left[1 + (-1)^x \sin 2nx \right]^{\frac{1}{2}} \checkmark$$

$$\text{Here } x = 8, \quad x = \pi, \quad n = \frac{1}{4}$$

$$u_8(\pi) = \left(\frac{1}{4} \right)^8 \left[1 + (-1)^8 \sin 2 \times \frac{1}{4} \pi \right]^{\frac{1}{2}}$$

$$= \left(\frac{1}{2} \right)^{16} \left[1 + \sin \frac{\pi}{2} \right]^{\frac{1}{2}}$$

$$= \left(\frac{1}{2} \right)^{16} \times (2)^{\frac{1}{2}} = \frac{1}{2^{16}} \times \frac{1}{2^{-\frac{1}{2}}}$$

$$u_8(\pi) = \left(\frac{1}{2} \right)^{3\frac{1}{2}}$$

Hence proved

(9) n^{th} derivative of $y = e^{ax} \sin(bx + c)$

$$y = e^{ax} \sin(bx + c)$$

$$y_1 = e^{ax} b \cos(bx + c) + e^{ax} a \sin(bx + c)$$

$$y_1 = e^{ax} [b \cos(bx + c) + a \sin(bx + c)]$$

Put $b = r \sin \theta$

$a = r \cos \theta$

$r = \sqrt{a^2 + b^2}$

$\theta = \tan^{-1} \frac{b}{a}$

$$y_1 = e^{ax} [r \sin \theta \cos(bx + c) + r \cos \theta \sin(bx + c)]$$

$$y_1 = r e^{ax} [\sin \theta \cos(bx + c) + \cos \theta \sin(bx + c)]$$

$$y_1 = r e^{ax} \sin(bx + c + \theta)$$

Similarly

$$y_2 = r^2 e^{ax} \sin(bx + c + 2\theta)$$

$$y_3 = r^3 e^{ax} \sin(bx + c + 3\theta)$$

$$y_n = r^n e^{an} \sin(bn + c + n\theta)$$

$$y_n = (a^2 + b^2)^{\frac{n}{2}} e^{an} \sin(bn + c + n \tan^{-1} \frac{b}{a})$$

If $a=1, b=1, c=0$

$$y = e^n \sin n$$

$$y_n = (2)^{\frac{n}{2}} e^n \sin\left(n + \frac{n\pi}{4}\right)$$

(10) n^{th} derivative of $y = e^{ax} \cos(bx + c)$

$$y = e^{ax} \cos(bx + c)$$

$$y_1 = e^{ax} (-b) \sin(bx + c) + e^{ax} a \cos(bx + c)$$

$$y_1 = e^{ax} [a \cos(bx + c) - b \sin(bx + c)]$$

Put $a = r \cos \theta$

$b = r \sin \theta$

$$r = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1} \frac{b}{a}$$

$$y_1 = e^{ax} [r \cos \theta \cos(bx + c) - r \sin \theta \sin(bx + c)]$$

$$y_1 = r e^{ax} [\cos \theta \cos(bx + c) - \sin \theta \sin(bx + c)]$$

$$y_1 = r e^{ax} \cos(bx + c + \theta)$$

Similarly

$$y_2 = r^2 e^{ax} \cos(bx + c + 2\theta)$$

$$y_3 = r^3 e^{ax} \cos(bx + c + 3\theta)$$

$\vdots$

$\vdots$

$$y_n = r^n e^{ax} \cos(bx + c + n\theta)$$

$$y_n = (a^2 + b^2)^{\frac{n}{2}} e^{ax} \cos(bx + c + n \tan^{-1} \frac{b}{a})$$

If $a=1$, $b=1$, $c=0$

$$y = e^x \cos x$$

$$y_n = (2)^{\frac{n}{2}} e^x \cos(x + n\frac{\pi}{4})$$

Q.5 Find the n^{th} derivative of $y = e^x \sin^3 x$

$$y = e^x \sin^3 x$$

$$\because \sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\Rightarrow \sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x)$$

$$y = \frac{e^x}{4} (3 \sin x - \sin 3x)$$

$$y = \frac{1}{4} \left[3 \underline{e^x \sin x} - \underline{e^x \sin 3x} \right]$$

$$y_n = \frac{1}{4} \left[3 \frac{d^n}{dx^n} e^x \sin x - \frac{d^n}{dx^n} e^x \sin 3x \right]$$

$$y_n = \frac{1}{4} \left[3 (1^2 + 1^2)^{\frac{n}{2}} e^x \sin(x + n \tan^{-1} 1) - (1^2 + 3^2)^{\frac{n}{2}} e^x \sin(3x + \tan^{-1} 3) \right]$$

$$e^x \sin(3x + \tan^{-1} 3)$$

$$y_n = \frac{3}{4} (2)^{\frac{n}{2}} e^x \sin\left(x + \frac{n\pi}{4}\right) - \frac{1}{4} (10)^{\frac{n}{2}} e^x \sin(3x + n \tan^{-1} 3)$$

Successive Differentiation (n^{th} order derivatives Formulae)

(1) n^{th} Derivative of $y = x^m$

$$y_n = m(m-1)(m-2)(m-3)\dots[m-(n-1)] x^{m-n}$$

If $m < n$

$$y_n = n!$$

If $m = n$

$$y_n = 0$$

If $m > n$

(2) n^{th} derivative of $y = (ax + b)^m$

$$y_n = m(m-1)(m-2)(m-3)\dots[m-(n-1)] (ax + b)^{m-n} \times a^n$$

If $m < n$

$$y_n = a^n n!$$

If $m = n$

$$y_n = 0$$

If $m > n$

(3) n^{th} derivative of $y = a^{mx}$

$$y_n = m^n a^{mx} (\log a)^n$$

If $m = 1$ then $y = a^x$

$$y_n = a^x (\log a)^n$$

(4) n^{th} derivative of $y = e^{mx}$

$$y_n = m^n e^{mx}$$

If $m = 1$ then $y = e^x$

$$y_n = e^x$$

Imp

(5) n^{th} derivative of $y = \frac{1}{ax + b}$

$$y_n = \frac{(-1)^n \times n! \times a^n}{(ax + b)^{n+1}}$$

If $a = 1$ and $b = 0$ then $y = \frac{1}{x}$

$$y_n = \frac{(-1)^n \times n!}{x^{n+1}}$$

(6) n^{th} derivative of $y = \log(ax + b)$

$$y_n = \frac{(-1)^{n-1} \times (n-1)! \times a^n}{(ax + b)^n}$$

If $a = 1$ and $b = 0$ then $y = \log x$

$$y_n = \frac{(-1)^{n-1} \times (n-1)!}{x^n}$$

imp

(7) n^{th} derivative of $y = \sin(ax + b)$

$$y_n = a^n \sin\left(ax + b + \frac{n\pi}{2}\right)$$

If $a = 1$ and $b = 0$ then $y = \sin x$

$$y_n = \sin\left(x + \frac{n\pi}{2}\right)$$

(8) n^{th} derivative of $y = \cos(ax + b)$

$$y_n = a^n \cos\left(ax + b + \frac{n\pi}{2}\right)$$

If $a = 1$ and $b = 0$ then $y = \cos x$

$$y_n = \cos\left(x + \frac{n\pi}{2}\right)$$

imp

(9) n^{th} derivative of $y = e^{ax} \sin(bx + c)$

$$y_n = (a^2 + b^2)^{\frac{n}{2}} e^{ax} \sin\left(bx + c + n \tan^{-1} \frac{b}{a}\right)$$

If $a = 1$, $b = 1$ and $c = 0$ then $y = e^x \sin x$

$$y_n = (2)^{\frac{n}{2}} e^x \sin\left(x + \frac{n\pi}{4}\right)$$

(10) n^{th} derivative of $y = e^{ax} \cos(bx + c)$

$$y_n = (a^2 + b^2)^{\frac{n}{2}} e^{ax} \cos\left(bx + c + n \tan^{-1} \frac{b}{a}\right)$$

If $a = 1$, $b = 1$ and $c = 0$ then $y = e^x \cos x$

$$y_n = (2)^{\frac{n}{2}} e^x \cos\left(x + \frac{n\pi}{4}\right)$$

DPP - 2

Topic : Successive Differentiation (Part-2)

Q.1 Find the n th derivative of $\log \sqrt{\frac{2x+1}{x-2}}$

Q.2 Find the n th derivative of $y = \sin^2 x \cos^2 x$

Q.3 Find the n th derivative of $y = \sin^2 x \cos^3 x$

Q.4 Find the n th derivative of $y = e^x \sin x \cos x$

Q.5 If $I_n = \frac{d^n}{dx^n}(x^n \log x)$, prove that $I_n = nI_{n-1} + (n-1)!$

AKTU 2017

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 3

Today's target

- Leibnitz Theorem (Part-1)
- PYQs
- DPP



By Gulshan sir

If $y = uv$

$$y_n = (n_{c_0} u_n v) + (n_{c_1} u_{n-1} v_1) + (n_{c_2} u_{n-2} v_2) + \dots + (n_{c_n} u v_n)$$

Where

- u and v are the function of x
- Generally $v = \text{polynomial } (x, x^2, x^3, x^2 + 2x + 1 \text{ etc})$
- $n_{c_r} = \frac{n!}{r!(n-r)!}$

$$n_{c_0} = 1$$

$$n_{c_1} = n$$

$$n_{c_2} = \frac{n(n-1)}{2}$$

$$n_{c_n} = 1$$

If $y = uv$

$$y_{n-1} = (n-1C_0 u_{n-1} v) + (n-1C_1 u_{n-2} v_1) + (n-1C_2 u_{n-3} v_2) + \dots$$

Where

- u and v are the function of x
- Generally $v = \text{polynomial } (x, x^2, x^3, x^2 + 2x + 1 \text{ etc})$
- $n_{C_r} = \frac{n!}{r!(n-r)!}$

$$n-1C_0 = 1$$

$$n-1C_1 = n-1$$

$$n-1C_2 = \frac{(n-1)(n-2)}{2}$$

$$n-1C_{n-1} = 1$$

$$\textcircled{y} \xrightarrow{n \text{ times differentiation}} \textcircled{y_n}$$

$$\textcircled{y_1} \xrightarrow{(n-1) \text{ times}} \underline{y_n}$$

$$\underline{y_1} \xrightarrow{n \text{ times}} \underline{y_{n+1}}$$

$$\text{//} \quad y_2 \xrightarrow{n \text{ times}} \underline{y_{n+2}}$$

$$y_2 \xrightarrow{(n-1) \text{ times}} y_{n+1}$$

$$y_2 \xrightarrow{(n-2) \text{ times}} y_n$$

$$y_3 \xrightarrow{n \text{ times}} y_{n+3}$$

Steps to solve the questions

Step – 1 : Try to write the given function as $u v$

Step – 2 : Find y_1 and simplify

(i) Try to remove denominator by taking LCM

(ii) Squaring both side to remove square root

(iii) Try to put y in RHS

Step – 3 : Find y_2 OR y_3

Step – 4 : Apply Leibnitz Theorem

Q.1 Find the n^{th} derivative of $x^2 \sin x$ at $x = 0$

$$y = \underbrace{x^2}_v \underbrace{\sin x}_u$$

Differentiating n times by Leibnitz theorem

$$y_n = {}^nC_0 u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_n u v_n$$

$$y_n = 1 \times \sin\left(x + \frac{n\pi}{2}\right) x^2 + n \sin\left\{x + (n-1)\frac{\pi}{2}\right\} (2x) + \frac{n(n-1)}{2} \sin\left\{x + (n-2)\frac{\pi}{2}\right\} x^2$$

$$y_n = x^2 \sin\left(x + \frac{n\pi}{2}\right) + 2nx \sin\left\{x + (n-1)\frac{\pi}{2}\right\} + n(n-1) \sin\left\{x + (n-2)\frac{\pi}{2}\right\}$$

$$y_n = 0 + 0 + n(n-1) \sin(n-2)\frac{\Delta}{2}$$

$$y_n = (n^2 - n) \sin\left(n\frac{\Delta}{2} - \pi\right)$$

$$y_n = (n^2 - n) \left\{ -\sin\left(\pi - n\frac{\Delta}{2}\right) \right\}$$

$$y_n = (n - n^2) \sin \frac{n\Delta}{2}$$

Q.2 Find the n^{th} derivative of $e^x \log x$

$$y = \underset{u}{e^x} \underset{v}{\log x}$$

Differentiating n times by Leibnitz Theorem

$$y_n = {}^nC_0 u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_n u v_n$$

$$y_n = 1 \times e^x \log x + n e^x \times \frac{1}{x} + \frac{n(n-1)}{2} e^x \times \left(-\frac{1}{x^2}\right) + \dots + \frac{1 \times e^x (-1)^{n-1} (n-1)!}{x^n}$$

$$y_n = e^x \left[\log x + \frac{n}{x} - \frac{n(n-1)}{2} \times \frac{1}{x^2} + \dots + \frac{(-1)^{n-1} (n-1)!}{x^n} \right]$$

Q.3 If $y = (x^2 - 1)^n$, prove that $(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$

$$y = (x^2 - 1)^n$$

$$y_1 = n(x^2 - 1)^{n-1} \times 2x$$

Multiply by $(x^2 - 1)$ on both sides

$$(x^2 - 1)y_1 = n(x^2 - 1)^{n-1}(x^2 - 1) \times 2x$$

$$(x^2 - 1)y_1 = n(x^2 - 1)^n 2x$$

$$(x^2 - 1)y_1 = 2nxy$$

$$(x^2 - 1)y_2 + y_1 2n = 2n(xy_1 + y)$$

$$(x^2 - 1)y_2 + 2ny_1 = 2nxy_1 + 2ny$$

$$(x^2 - 1)y_2 + 2ny_1 - 2nxy_1 - 2ny = 0$$

$$(x^2 - 1)y_2 + (2 - 2n)xy_1 - 2ny = 0$$

Differentiating n times by Leibnitz
Theorem

$$\left[{}^nC_0 y_{n+2} (n^2-1) + {}^nC_1 y_{n+1} (2n) + {}^nC_2 y_n (2) \right] + (2-2n) \left[{}^nC_0 y_{n+1} (x) + {}^nC_1 y_n (1) \right] - 2n y_n = 0$$

$$(n^2-1) y_{n+2} + \underline{2nx y_{n+1}} + \underline{\frac{n(n-1)}{2} y_n \times 2} + \underline{(2-2n) y_{n+1} (x)} + \underline{(2-2n) n y_n} - \underline{2n y_n} = 0$$

$$(n^2-1) y_{n+2} + (2n\cancel{x} + 2n - 2n\cancel{x}) y_{n+1} + (n^2 - n + 2\cancel{n} - 2n^2 - 2\cancel{n}) y_n = 0$$

$$(n^2-1) y_{n+2} + 2n y_{n+1} + (-n^2 - n) y_n = 0$$

$$(n^2-1) y_{n+2} + 2n y_{n+1} - n(n+1) y_n = 0 \quad \text{Hence proved}$$

prove that $(1+x^2)y_{n+2} + [(2n+2)x-1]y_{n+1} + n(n+1)y = 0$

AKTU 2018, 2021

$$y = e^{\tan^{-1} x}$$

$$y_1 = e^{\tan^{-1} x} \times \frac{1}{1+x^2}$$

$$(1+x^2)y_1 = e^{\tan^{-1} x}$$

$$(1+x^2)y_1 = y$$

$$(1+x^2)y_2 + y_1 \times 2x = y_1 (1+x^2)y_{n+2} + 2nny_{n+1} + \frac{n(n-1)}{x} y_n \times x + (2n-1)y_{n+1} + 2ny_n = 0$$

$$(1+x^2)y_2 + 2xy_1 - y_1 = 0$$

$$(1+x^2)y_2 + (2n-1)y_1 = 0$$

Differentiating n times by Leibnitz Theorem

$$\left[\binom{n}{0} y_{n+2} (1+x^2) + \binom{n}{1} y_{n+1} (2x) + \binom{n}{2} y_n (2) \right] + \left[\binom{n}{0} y_{n+1} (2n-1) + \binom{n}{1} y_n (2) \right] = 0$$

$$(1+n^2)y_{n+2} + (2nn+2n-1)y_{n+1} + (n^2-n+2n)y_n = 0$$

$$(1+n^2)y_{n+2} + [(2n+2)n-1]y_{n+1} + (n^2+n)y_n = 0$$

$$\Rightarrow (1+n^2)y_{n+2} + [(2n+2)n-1]y_{n+1} + n(n+1)y_n = 0$$

Q.5 If $y = a \cos(\log x) + b \sin(\log x)$, prove that $x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$

$$y = a \cos(\log n) + b \sin(\log n)$$

$$y_1 = -a \sin(\log n) \times \frac{1}{n} + b \cos(\log n) \times \frac{1}{n}$$

$$y_1 = \frac{-a \sin(\log n) + b \cos(\log n)}{n}$$

$$ny_1 = -a \sin(\log n) + b \cos(\log n)$$

$$ny_2 + y_1 = -a \cos(\log n) \times \frac{1}{n} - b \sin(\log n) \times \frac{1}{n}$$

$$ny_2 + y_1 = \frac{-\{a \cos(\log n) + b \sin(\log n)\}}{n}$$

$$n(ny_2 + y_1) = -y$$

$$n^2 y_2 + ny_1 = -y$$

$$n^2 y_2 + ny_1 + y = 0$$

$$n^2 y'' + n y' + y = 0$$

Differentiating n times by Leibnitz theorem

$$\left[nC_0 y_{n+2}^{(n^2)} + nC_1 y_{n+1}^{(2n)} + nC_2 y_n^{(2)} \right] + \left[nC_0 y_{n+1}^{(n)} + nC_1 y_n^{(1)} \right] + y_n = 0$$

$$n^2 y_{n+2} + 2nny_{n+1} + \frac{n(n-1)}{2} y_n \times 2 + ny_{n+1} + ny_n + y_n = 0$$

$$n^2 y_{n+2} + (2n+1)ny_{n+1} + (n^2 - n + n + 1)y_n = 0$$

$$n^2 y_{n+2} + (2n+1)ny_{n+1} + (n^2 + 1)y_n = 0 \quad \text{Henu proved}$$

Q.6 If $y = \sin \log_e(x^2 + 2x + 1)$

prove that $(1+x)^2 y_{n+2} + (2n+1)(1+x)y_{n+1} + (n^2+4)y_n = 0$

AKTU 2019

$$y = \sin \log_e(x^2 + 2x + 1)$$

$$y_1 = \cos \log_e(x^2 + 2x + 1) \times \frac{1}{x^2 + 2x + 1} (2x + 2)$$

$$y_1 = \cos \log_e(x^2 + 2x + 1) \times \frac{2(x+1)}{(x+1)^2}$$

$$(n+1)y_1 = 2 \cos \log_e(x^2 + 2x + 1)$$

$$(n+1)y_2 + y_1 = -2 \sin \log_e(x^2 + 2x + 1) \times \frac{1 \times (2n+2)}{(x^2 + 2x + 1)}$$

$$(n+1)y_2 + y_1 = \frac{-2y \times 2(n+1)}{(n+1)^2}$$

$$(n+1)^2 y_2 + (n+1)y_1 = -4y$$

$$(n+1)^2 y_2 + (n+1)y_1 + 4y = 0$$

$$(n+1)^2 y_{n+2} + (n+1) y_{n+1} + 4y_n = 0$$

$$\frac{2nn+2n+n+1}{2n(n+1)+1(n+1)} \\ (n+1)(2n+1)$$

Differentiating n times by Leibnitz theorem

$$\left[nC_0 y_{n+2} (n+1)^2 + nC_1 y_{n+1} 2(n+1) + nC_2 y_n (2) \right] + \left[nC_0 y_{n+1} (n+1) + nC_1 y_n (1) \right] + 4y_n = 0$$

$$(n+1)^2 y_{n+2} + 2n(n+1) y_{n+1} + \frac{n(n-1)}{2} y_n \times 2 + (n+1) y_{n+1} + n y_n + 4y_n = 0$$

$$(n+1)^2 y_{n+2} + (2nn+2n+n+1) y_{n+1} + (n^2 - \cancel{n} + \cancel{n} + 4) y_n = 0$$

$$(n+1)^2 y_{n+2} + (2n+1)(n+1) y_{n+1} + (n^2+4) y_n = 0$$

Hence proved

DPP - 3

Topic : Leibnitz Theorem (Part-1)

Q.1 If $y = x^2 e^{2x}$ determine $(y_n)_0$

Q.2 $y = \tan^{-1} \left(\frac{a+x}{a-x} \right)$, prove that $(a^2 + x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$

Q.3 If $\cos^{-1} \left(\frac{y}{b} \right) = \log \left(\frac{x}{m} \right)^m$, prove that $x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2 + m^2)y_n = 0$

Q.4 If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, prove that $(1-x^2)y_{n+2} - (2n+3)xy_{n+1} - (n+1)^2 y_n = 0$

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 4

Today's target

- Leibnitz Theorem (Part-2)
- PYQs
- DPP



By Gulshan sir

Q.1 If $y^{\frac{1}{m}} + y^{\frac{-1}{m}} = 2x$, prove that $(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$

AKTU2015

$$y^{\frac{1}{m}} + y^{\frac{-1}{m}} = 2x$$

Put $y^{\frac{1}{m}} = t$

$$t + t^{-1} = 2x$$

$$t + \frac{1}{t} = 2x$$

$$\frac{t^2 + 1}{t} = 2x$$

$$t^2 + 1 = 2xt$$

$$t^2 - 2xt + 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{2x \pm \sqrt{4x^2 - 4}}{2}$$

$$t = \frac{2x \pm 2\sqrt{x^2 - 1}}{2}$$

$$t = \cancel{2} \frac{(x \pm \sqrt{x^2 - 1})}{\cancel{2}}$$

$$t = x \pm \sqrt{x^2 - 1}$$

Taking +ve sign

$$t = x + \sqrt{x^2 - 1}$$

$$y^{\frac{1}{m}} = x + \sqrt{x^2 - 1}$$

$$y = [x + \sqrt{x^2 - 1}]^m$$

$$y_1 = m[x + \sqrt{x^2 - 1}]^{m-1} \times \left(1 + \frac{1 \times 2x}{2\sqrt{x^2 - 1}}\right)$$

$$y_1 = m(x + \sqrt{x^2 - 1})^{m-1} \times \frac{(x + \sqrt{x^2 - 1})'}{\sqrt{x^2 - 1}}$$

$$y_1 = \frac{m(x + \sqrt{x^2 - 1})^m}{\sqrt{x^2 - 1}}$$

$$\sqrt{x^2 - 1} y_1 = my$$

Squaring both side

$$(x^2 - 1)y_1^2 = m^2 y^2$$

$$(x^2 - 1) \times 2y_1 y_2 + y_1^2 2x = 2m^2 y y_1$$

$$2y_1 [(x^2 - 1)y_2 + xy_1] = 2y_1 (my)$$

$$(x^2 - 1)y_2 + xy_1 = m^2 y$$

Differentiating n times by Leibnitz Theorem

$$\left[{}^nC_0 y_{n+2} (x^2 - 1) + {}^nC_1 y_{n+1} (2x) + {}^nC_2 y_n (2) \right] + \left[{}^nC_0 y_{n+1} x + {}^nC_1 y_n (1) \right]$$

$$(x^2 - 1)y_{n+2} + 2nxy_{n+1} + \frac{2n(n-1)}{2}y_n + xy_{n+1} + ny_n = m^2 y_n$$

$$(x^2 - 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - n + n - m^2)y_n = 0$$

$$(x^2 - 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0$$

Q.2 If $y = (\sinh^{-1}x)^2$, prove that $(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2y_n = 0$

$$y = (\sinh^{-1}x)^2$$

$$\frac{d}{dx}(\sinh^{-1}x) = \frac{1}{\sqrt{x^2+1}}$$

$$y_1 = 2(\sinh^{-1}x) \times \frac{1}{\sqrt{x^2+1}}$$

$$y_1 \sqrt{x^2+1} = 2(\sinh^{-1}x)$$

Squaring both side

$$y_1^2 (x^2+1) = 4(\sinh^{-1}x)^2$$

$$y_1^2 (n^2+1) = 4y$$

$$y_1^2 (2n) + (n^2+1) 2y_1 y_2 = 4y_1$$

$$2y_1 [ny_1 + (n^2+1)y_2] = 2y_1 (2)$$

$$(n^2+1)y_2 + ny_1 = 2$$

AKTU2015

$$(x^2+1)y_2'' + xy_1' = 2$$

Differentiating n times by Leibnitz theorem

$$\left[{}^nC_0 y_{n+2} (x^2+1) + {}^nC_1 y_{n+1} (2x) + {}^nC_2 y_n (2) \right] + \left[{}^nC_0 y_{n+1} x + {}^nC_1 y_n (1) \right] = 0$$

$$(x^2+1)y_{n+2} + 2nxy_{n+1} + \frac{2n(n-1)}{2} y_n + ny_{n+1} + ny_n = 0$$

$$(x^2+1)y_{n+2} + (2n+1)y_{n+1} + (n^2-n+n)y_n = 0$$

$$(x^2+1)y_{n+2} + (2n+1)y_{n+1} + n^2 y_n = 0$$

AKTU 2023

$$y\sqrt{n^2-1} = \log(n + \sqrt{n^2-1})$$

$$y \times \frac{1 \times \cancel{x}^n}{\cancel{x}^n \sqrt{n^2-1}} + \sqrt{n^2-1} y_1 = \frac{1}{(n + \sqrt{n^2-1})} \times \left(1 + \frac{1 \times \cancel{x}^n}{\cancel{x}^n \sqrt{n^2-1}}\right)$$

$$\frac{ny + (n^2-1)y_1}{\sqrt{n^2-1}} = \frac{1}{(n + \sqrt{n^2-1})} \times \left(\frac{\sqrt{n^2-1} + n}{\sqrt{n^2-1}}\right)$$

$$\frac{ny + (n^2-1)y_1}{\sqrt{n^2-1}} = \frac{1}{\sqrt{n^2-1}}$$

$$(n^2-1)y_1 + ny = 1$$

$$(x^2-1)y'' + xy' = 1$$

Differentiating n times by Leibnitz theorem

$$\left[{}^nC_0 y^{(n+1)} (x^2-1) + {}^nC_1 y^{(n)} (2x) + {}^nC_2 y^{(n-1)} (2) \right] + \left[{}^nC_0 y^{(n)} + {}^nC_1 y^{(n-1)} \right] = 0$$

$$(x^2-1)y_{n+1} + 2nx y_n + \frac{2n(n-1)}{2} y_{n-1} + x y_n + n y_{n-1} = 0$$

$$(x^2-1)y_{n+1} + (2n+1)x y_n + (n^2-n+n)y_{n-1} = 0$$

$$(x^2-1)y_{n+1} + (2n+1)x y_n + n^2 y_{n-1} = 0$$

Q.4 If $y = x^{n-1} \log x$, then find y_n

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$$y = x^{n-1} \log x$$

$$y_1 = x^{n-1} \cdot \frac{1}{x} + \log x \cdot (n-1) x^{n-2}$$

$$y_1 = \frac{x^{n-1} + (n-1) x^{n-2} x \log x}{x}$$

$$x y_1 = x^{n-1} + (n-1) x^{n-1} \log x$$

$$x y_1 = x^{n-1} + (n-1) y$$

Differentiating $(n-1)$ times by Leibnitz theorem

$${}^{n-1}C_0 y_n + {}^{n-1}C_1 y_{n-1} = (n-1)! + (n-1) y_{n-1}$$

$$x y_n + (n-1) y_{n-1} = (n-1)! + (n-1) y_{n-1}$$

$$x y_n = (n-1)!$$

$$y_n = \frac{(n-1)!}{x}$$



Q.5 If $y = \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$, prove that $(1-x^2)y_n - [2(n-1)x+1]y_{n-1} - (n-1)(n-2)y_{n-2} = 0$

$$y = \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$$

Taking log both side

$$\log y = \log \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$$

$$\log y = \frac{1}{2} \log \left(\frac{1+x}{1-x}\right)$$

$$2 \log y = \log(1+x) - \log(1-x)$$

$$2 \times \frac{1}{y} y_1 = \frac{1}{1+x} + \frac{1}{1-x}$$

$$\frac{2y_1}{y} = \frac{1-x + 1+x}{(1+x)(1-x)}$$

$$\frac{2y_1}{y} = \frac{2}{1-x^2}$$

$$(1-x^2)y_1 = y$$

$$(1-x^2)y_1 = y$$

Differentiating $(n-1)$ times by Leibnitz theorem

$${}^{n-1}C_0 y_n (1-x^2) + {}^{n-1}C_1 y_{n-1} (-2x) + {}^{n-1}C_2 y_{n-2} (-2) = y_{n-1}$$

$$(1-x^2)y_n - 2(n-1)x y_{n-1} - \frac{2(n-1)(n-2)}{2} y_{n-2} - y_{n-1} = 0$$

$$(1-x^2)y_n - [2(n-1)x + 1] y_{n-1} - (n-1)(n-2) y_{n-2} = 0$$

Hence proved

DPP - 4

Topic : Leibnitz Theorem (Part-2)

Q.1 If $y = (1 - x)^{-\alpha} e^{-\alpha x}$, prove that $(1 - x)y_{n+1} - (n + \alpha x)y_n - n\alpha y_{n-1} = 0$

Q.2 If $x = \tan(\log y)$, prove that $(1 + x^2)y_{n+1} - (2nx - 1)y_n + n(n - 1)y_{n-1} = 0$

AKTU 2014

Q.3 If $x = \cosh\left(\frac{1}{m} \log y\right)$, prove that $(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$

Q.4 If $y = x^n \log x$, prove that (i) $y_{n+1} = \frac{n!}{x}$ (ii) $x^2 y_{p+2} + (2p - 2n + 1)xy_{p+1} + (p - n)^2 y_p = 0$

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 5

Today's target

- Leibnitz Theorem (Part-3)
- PYQs
- DPP



By Gulshan sir

Determination of n^{th} derivative of a function at $x = 0$

Steps to solve the questions

Step – 1 : Write the given function as $y = f(x)$ _____ ①

Step – 2 : Find y_1 and simplify _____ ②

(i) Try to remove denominator by taking LCM

(ii) Squaring both side to remove square root

(iii) Try to put y in RHS

Step – 3 : Find y_2 _____ ③

Step – 4 : Apply Leibnitz Theorem _____ ④

Step – 5 : Put $x = 0$ in equations of (1), (2), (3) and (4). _____ ⑤

Step – 6 : On putting $x = 0$, two cases arises

(i) when $n = \text{odd}$

(ii) when $n = \text{even integer.}$

Q.1 If $y = \sin(m \sin^{-1} x)$, find $y_n(0)$

AKTU 2018, 2019, 2021

$$y = \sin(m \sin^{-1} x) \quad \text{--- (1)}$$

$$y_1 = \cos(m \sin^{-1} x) \times \frac{m}{\sqrt{1-x^2}} \quad \text{--- (2)}$$

$$(\sqrt{1-x^2}) y_1 = m \cos(m \sin^{-1} x)$$

Squaring both side

$$(1-x^2) y_1^2 = m^2 \cos^2(m \sin^{-1} x)$$

$$(1-x^2) y_1^2 = m^2 [1 - \sin^2(m \sin^{-1} x)]$$

$$(1-x^2) y_1^2 = m^2 (1-y^2)$$

$$(1-x^2) 2y_1 y_2 + y_1^2 (-2x) = m^2 (-2y y_1)$$

$$2y_1 [(1-x^2) y_2 - x y_1] = 2y_1 (-m^2 y)$$

$$(1-x^2) y_2 - x y_1 + m^2 y = 0 \quad \text{--- (3) } \checkmark$$

Differentiating n times by Leibnitz theorem

$${}^n C_0 y (1-x^2) + {}^n C_1 y (-2x) + {}^n C_2 y (-2) - \left[{}^n C_0 y x + {}^n C_1 y (1) \right] + m^2 y = 0$$

$$(1-x^2) y_{n+2} - 2n x y_{n+1} - \frac{n(n-1)}{2} y_n - x y_{n+1} - n y_n + m^2 y_n = 0$$

$$(1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 - n + m^2) y_n = 0$$

$$(1-n^2)y_{n+2} - (2n+1)ny_{n+1} - (n^2-m^2)y_n = 0$$

④

Put $n=0$ in ①, ②, ③ and ④

$$y(0) = 0$$

$$y_1(0) = m$$

$$y_2(0) = -m^2 y(0) = 0$$

$$y_{n+2}(0) = (n^2-m^2)y_n(0)$$

⑤

Put $n=1, 2, 3, 4, 5, \dots$ ⑤

$$y_3(0) = (1^2-m^2)y_1(0) = m(1^2-m^2)$$

$$y_4(0) = (2^2-m^2)y_2(0) = 0$$

$$y_5(0) = (3^2-m^2)y_3(0) = m(1^2-m^2)(3^2-m^2)$$

$$y_6(0) = (4^2-m^2)y_4(0) = 0$$

$$\vdots$$

$$y_n(0) = \begin{cases} 0 & \text{When } n \text{ is even} \\ m(1^2-m^2)(3^2-m^2) \dots [(n-2)^2-m^2] & \text{When } n \text{ is odd} \end{cases}$$

Q. 2 If $y = (\sin^{-1} x)^2$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$ Also find the value of n^{th} derivative of y for $x = 0$

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$$y = (\sin^{-1} x)^2 \quad \text{--- (1)}$$

$$y_1 = 2 \sin^{-1} x \times \frac{1}{\sqrt{1-x^2}} \quad \text{--- (2)}$$

$$y_1 \sqrt{1-x^2} = 2 \sin^{-1} x$$

Squaring both side

$$(1-x^2)y_1^2 = 4(\sin^{-1} x)^2$$

$$(1-x^2)y_1^2 = 4y$$

$$(1-x^2) \times 2y_1 y_2 + y_1^2 (-2x) = 4y_1$$

$$2y_1 [(1-x^2)y_2 - xy_1] = 2y_1 \times 2$$

$$(1-x^2)y_2 - xy_1 = 2 \quad \text{--- (3)}$$

Differentiating n times by Leibnitz theorem

$$n {}^nC_0 y_{n+2} (1-x^2) + n {}^nC_1 y_{n+1} (-2x) + n {}^nC_2 y_n (-2) - \left[n {}^nC_0 y_{n+1} (x) + n {}^nC_1 y_n (1) \right]$$

$$= 0$$

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{2n(n-1)}{2}y_n - xy_{n+1} - ny_n = 0 \quad \text{Put } n=1, 2, 3, 4 \dots \text{ in (5)}$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - \cancel{n} + \cancel{n})y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0 \quad \text{--- (4)}$$

Put $n=0$, in (1), (2), (3), (4)

$$y(0) = 0$$

$$y_1(0) = 0$$

$$y_2(0) = 2$$

$$y_{n+2}(0) = n^2 y_n(0) \quad \text{--- (5)}$$

$$y_3(0) = 1^2 y_1(0) = 0$$

$$y_4(0) = 2^2 y_2(0) = 2 \times 2^2$$

$$y_5(0) = 3^2 y_3(0) = 0$$

$$y_6(0) = 4^2 y_4(0) = 2 \times 2^2 \times 4^2$$

⋮

⋮

$$y_n(0) = \begin{cases} 0 & \text{When } n \text{ is odd} \\ 2 \times 2^2 \times 4^2 \times \dots \times (n-2)^2 & \text{When } n \text{ is even} \end{cases}$$

Q.3 If $y = e^{m \cos^{-1} x}$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$ and calculate $y_n(0)$

OR

$$\log y = x$$

AKTU 2020

$\cos^{-1} x = \log(y)^{\frac{1}{m}}$, then show $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$ and hence

calculate y_n when $x = 0$

AKTU 2016

$$\cos^{-1} x = \log(y)^{\frac{1}{m}}$$

$$\cos^{-1} x = \frac{1}{m} \log y$$

$$m \cos^{-1} x = \log y$$

$$\log y = m \cos^{-1} x$$

$$y = e^{m \cos^{-1} x}$$

$$y = e^{m \cos^{-1} x} \quad \text{--- (1)}$$

$$y_1 = e^{m \cos^{-1} x} \times \frac{(-m)}{\sqrt{1-x^2}} \quad \text{--- (2)}$$

$$(\sqrt{1-x^2}) y_1 = -m y$$

Squaring both side

$$(1-x^2) y_1^2 = m^2 y^2$$

$$(1-x^2) x^2 y_1 y_2 + y_1^2 (-2x) = m^2 x^2 y_1 y_2$$

$$2 y_1 [(1-x^2) y_2 - x y_1] = 2 y_1 (m^2 y)$$

$$(1-x^2) y_2 - x y_1 = m^2 y$$

$$(1-x^2)y_2 - xy_1 = m^2 y \quad \text{--- (3)}$$

Differentiating n times by Leibnitz Theorem

$$n C_0 y_{n+2} (1-x^2) + n C_1 y_{n+1} (-2x) + n C_2 y_n (-2) - \left[n C_0 y_{n+1} (x) + n C_1 y_n (1) \right] = m^2 y_n$$

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2}y_n - ny_{n+1} - ny_n - m^2 y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - \cancel{n} + \cancel{n} + m^2)y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + m^2)y_n = 0 \quad \text{--- (4)}$$

Put $n=0$ in (1) (2) (3) (4)

$$y(0) = e^{\frac{m\pi}{2}}$$

$$y_1(0) = -m e^{\frac{m\pi}{2}}$$

$$y_2(0) = m^2 y(0) = m^2 e^{\frac{m\pi}{2}}$$

$$y_{n+2}(0) = (n^2 + m^2) y_n(0)$$

--- (5)

Put $n=1, 2, 3, 4, \dots$ in (5)

$$y_3(0) = (1^2 + m^2)y_1(0) = -m e^{\frac{m\pi}{2}} (1^2 + m^2) \checkmark$$

$$y_4(0) = (2^2 + m^2)y_2(0) = m^2 e^{\frac{m\pi}{2}} (2^2 + m^2)$$

$$y_5(0) = (3^2 + m^2)y_3(0) = -m e^{\frac{m\pi}{2}} (1^2 + m^2)(3^2 + m^2) \checkmark$$

$$y_6(0) = (4^2 + m^2)y_4(0) = m^2 e^{\frac{m\pi}{2}} (2^2 + m^2)(4^2 + m^2)$$

⋮

⋮

$$y_n(0) = \begin{cases} -m e^{\frac{m\pi}{2}} (1^2 + m^2)(3^2 + m^2) \dots [(n-2)^2 + m^2] & \text{When } n \text{ is odd} \\ m^2 e^{\frac{m\pi}{2}} (1^2 + m^2)(3^2 + m^2) \dots [(n-2)^2 + m^2] & \text{When } n \text{ is even} \end{cases}$$

Q.4 If $y = [(x + \sqrt{1+x^2})]^m$, find $y_n(0)$

$$y = [x + \sqrt{1+x^2}]^m \quad \text{--- (1)}$$

Taking log both side

$$\log y = \log [x + \sqrt{1+x^2}]^m$$

$$\log y = m \log [x + \sqrt{1+x^2}]$$

$$\frac{1}{y} y_1 = m \times \frac{1}{(x + \sqrt{1+x^2})} \times \left(1 + \frac{1 \times 2x}{2\sqrt{1+x^2}}\right)$$

$$\frac{1}{y} y_1 = \frac{m}{x + \sqrt{1+x^2}} \times \left(\frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}}\right)$$

AKTU 2018 2022

$$y_1 = \frac{my}{\sqrt{1+x^2}} \quad \text{--- (2)}$$

$$(\sqrt{1+x^2}) y_1 = my$$

Squaring both side

$$(1+x^2) y_1^2 = m^2 y^2$$

$$(1+x^2) 2y y_2 + y_1^2 (2x) = m^2 2y y_1$$

$$2y_1 [(1+x^2) y_2 + x y_1] = 2y_1 (m^2 y)$$

$$(1+x^2) y_2 + x y_1 = m^2 y$$

$$(1+x^2)y_2 + xy_1 - m^2y = 0 \text{ --- (3)}$$

Differentiating n times by Leibnitz theorem

$${}^nC_0 y (1+x^2) + {}^nC_1 y (2x) + {}^nC_2 y (2) + {}^nC_0 y (x) + {}^nC_1 y (1) - m^2 y_n = 0$$

$$(1+x^2)y_{n+2} + 2nxy_{n+1} + \frac{2n(n-1)}{2}y_n + xy_{n+1} + ny_n - m^2y_n = 0$$

$$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - n + n - m^2)y_n = 0$$

$$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0 \text{ --- (4)}$$

Put $n=0$ in eqn (1), (2), (3), (4)

$$y(0) = 1$$

$$y_1(0) = m y(0) = m$$

$$y_2(0) = m^2 y(0) = m^2$$

$$y_{n+2}(0) = (m^2 - n^2) y_n(0)$$

⑤

Put $n = 1, 2, 3, 4, \dots$
in ⑤

$$y_3(0) = (m^2 - 1^2) y_1(0) = m(m^2 - 1^2)$$

$$y_4(0) = (m^2 - 2^2) y_2(0) = m^2(m^2 - 2^2)$$

$$y_5(0) = (m^2 - 3^2) y_3(0) = m(m^2 - 1^2)(m^2 - 3^2)$$

$$y_6(0) = (m^2 - 4^2) y_4(0) = m^2(m^2 - 2^2)(m^2 - 4^2)$$

$$y_n(0) = \begin{cases} m(m^2 - 1^2)(m^2 - 3^2) \dots [m^2 - (n-2)^2] & \text{When } n \text{ is odd} \\ m^2(m^2 - 2^2)(m^2 - 4^2) \dots [m^2 - (n-2)^2] & \text{When } n \text{ is even} \end{cases}$$

$$\log y = \tan^{-1} x$$

$$y = e^{\tan^{-1} x}$$

DPP - 5

Topic : Leibnitz Theorem (Part-3)

Q.1 If $y = \cos(m \sin^{-1} x)$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - m^2)y_n = 0$ hence find $y_n(0)$

AKTU 2024

Q.2 If $y = e^{a \sin^{-1} x}$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + a^2)y_n = 0$ and calculate $y_n(0)$

Q.3 If $\log y = \tan^{-1} x$, show that $(1 + x^2)y_{n+2} + [2(n + 1)x - 1]y_{n+1} + n(n + 1)y_n = 0$ and hence find

y_3, y_4 , and y_5 for $x = 0$

Q.4 If $y = \tan^{-1} x$, prove that $(1 + x^2)y_{n+1} + 2nxy_n + n(n - 1)y_{n-1} = 0$ hence determine the values of all

derivatives of y with respect to x when $x = 0$

Q.5 If $y = [\log(x + \sqrt{1 + x^2})]^2$ find all the derivatives of y w.r.t x when $x = 0$

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 6

Today's target

- Partial Differentiation (Part-1)
- PYQs
- DPP



By Gulshan sir

Differentiation

Ordinary differentiation

One independent variable in the given function

$$y = f(x)$$

↑
DV
↑
IV

Partial differentiation

Two or more independent variable in the given function

$$z = f(x, y)$$

↑
DV
↑
IV
↑
IV

Partial differentiation

The process of finding differentiation of a function with respect to one independent variable and treating other independent variable as constant is called partial differentiation

Let $z = f(x, y)$

Partial Derivative of first order

$$(i) p = \frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x \quad (y \text{ is constant})$$

$$(ii) q = \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y \quad (x \text{ is constant})$$

Partial Derivative of second order

$$(i) r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 f}{\partial x^2} = z_{xx} = f_{xx}$$

$$(ii) s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 f}{\partial x \partial y} = z_{xy} = f_{xy}$$

$$(iii) t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 f}{\partial y^2} = z_{yy} = f_{yy}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Q.1 Find the first and second partial derivatives of $z = x^3 + y^3 - 3axy$

$$z = x^3 + y^3 - 3axy$$

Partially Diff. wrt x

$$\frac{\partial z}{\partial x} = 3x^2 + 0 - 3ay$$

$$\boxed{\frac{\partial z}{\partial x} = 3x^2 - 3ay}$$

Partially Diff. wrt y

$$\frac{\partial z}{\partial y} = 0 + 3y^2 - 3ax$$

$$\boxed{\frac{\partial z}{\partial y} = 3y^2 - 3ax}$$

$$\frac{\partial^2 z}{\partial x^2} = 3 \times 2x - 0$$

$$\boxed{\frac{\partial^2 z}{\partial x^2} = 6x}$$

$$\frac{\partial^2 z}{\partial y^2} = 3 \times 2y - 0$$

$$\boxed{\frac{\partial^2 z}{\partial y^2} = 6y}$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)$$

$$= \frac{\partial}{\partial y} (3x^2 - 3ay)$$

$$= -3a$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)$$

$$= \frac{\partial}{\partial x} (3y^2 - 3ax)$$

$$= -3a$$

Q.2 If $u = x^2y + y^2z + z^2x$ show that $u_x + u_y + u_z = (x + y + z)^2$

$$u = x^2y + y^2z + z^2x$$

$$u_x = \frac{\partial u}{\partial x} = 2xy + 0 + z^2$$

$$u_x = 2xy + z^2 \quad \text{--- (1)}$$

$$u_y = \frac{\partial u}{\partial y} = x^2 + 2yz + 0$$

$$u_y = x^2 + 2yz \quad \text{--- (2)}$$

$$u_z = \frac{\partial u}{\partial z} = 0 + y^2 + 2zx$$

$$u_z = y^2 + 2xz \quad \text{--- (3)}$$

Adding eqn (1), (2) and (3)

$$u_x + u_y + u_z = 2xy + z^2 + x^2 + 2yz + y^2 + 2xz$$

$$u_x + u_y + u_z = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$u_x + u_y + u_z = (x + y + z)^2$$

Hence proved

GW Q.3 If $z = e^{ax+by} f(ax - by)$, show that $b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$.

$$z = e^{ax+by} f(ax-by)$$

$$\frac{\partial z}{\partial x} = e^{ax+by} \times f'(ax-by) \times a + f(ax-by) \times e^{ax+by} \times a$$

$$b \frac{\partial z}{\partial x} = e^{ax+by} \times f'(ax-by) ab + e^{ax+by} \times f(ax-by) ab \quad \text{①}$$

$$\frac{\partial z}{\partial y} = e^{ax+by} \times f'(ax-by) (-b) + f(ax-by) e^{ax+by} \times b$$

$$a \frac{\partial z}{\partial y} = -e^{ax+by} \times f'(ax-by) ab + e^{ax+by} \times f(ax-by) ab \quad \text{②}$$

Adding eqⁿ ① and ②

$$b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2 e^{ax+by} f(ax-by) ab$$

$$b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$$

Henu proved

Q. 4 If $u(x, y, z) = \log(\tan x + \tan y + \tan z)$,

$$u = \log(\tan x + \tan y + \tan z)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z} \sec^2 x$$

$$\sin 2x \frac{\partial u}{\partial x} = \frac{\sin 2x \times \sec^2 x}{\tan x + \tan y + \tan z}$$

$$\sin 2x \frac{\partial u}{\partial x} = \frac{2 \sin x \cos x \times \frac{1}{\cos^2 x}}{\tan x + \tan y + \tan z}$$

show that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$.

$$\sin 2x \frac{\partial u}{\partial x} = \frac{2 \tan x}{\tan x + \tan y + \tan z} \quad \text{--- (1)}$$

Similarly

$$\sin 2y \frac{\partial u}{\partial y} = \frac{2 \tan y}{\tan x + \tan y + \tan z} \quad \text{--- (2)}$$

$$\sin 2z \frac{\partial u}{\partial z} = \frac{2 \tan z}{\tan x + \tan y + \tan z} \quad \text{--- (3)}$$

Adding (1), (2) and (3)

$$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$$

Q.5 If $z = \log(e^x + e^y)$, show that $rt - s^2 = 0$, where $r = \frac{\partial^2 z}{\partial x^2}$, $t = \frac{\partial^2 z}{\partial y^2}$, $s = \frac{\partial^2 z}{\partial x \partial y}$

$$z = \log(e^x + e^y)$$

$$\frac{\partial z}{\partial x} = \frac{1 \times e^x}{e^x + e^y}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{(e^x + e^y)e^x - e^x e^x}{(e^x + e^y)^2}$$

$$r = \frac{e^x(e^x + e^y - e^x)}{(e^x + e^y)^2}$$

$$r = \frac{e^x \times e^y}{(e^x + e^y)^2}$$

$$r = \frac{e^{x+y}}{(e^x + e^y)^2} \quad \text{--- (1)}$$

$$\frac{\partial z}{\partial y} = \frac{1 \times e^y}{e^x + e^y}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{(e^x + e^y)e^y - e^y e^y}{(e^x + e^y)^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{e^y(e^x + e^y - e^y)}{(e^x + e^y)^2}$$

$$t = \frac{e^{x+y}}{(e^x + e^y)^2} \quad \text{--- (2) AKTU 2015}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{e^y}{e^x + e^y} \right)$$

$$= \frac{(e^x + e^y)0 - e^y e^x}{(e^x + e^y)^2}$$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{e^{x+y}}{(e^x + e^y)^2}$$

$$s = -\frac{e^{x+y}}{(e^x + e^y)^2}$$

↳ (3)

$$rt - s^2 = \frac{e^{x+y}}{(e^x + e^y)^2} \times \frac{e^{x+y}}{(e^x + e^y)^2} - \frac{(e^{x+y})^2}{(e^x + e^y)^4}$$

~~$$rt - s^2 = \frac{(e^{x+y})^2}{(e^x + e^y)^4} - \frac{(e^{x+y})^2}{(e^x + e^y)^4}$$~~

$$rt - s^2 = 0$$

Hence proved

Q.6 If $x^2 = au + bv$, $y^2 = au - bv$ prove that $\left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial x}{\partial u}\right)_v = \frac{1}{2} = \left(\frac{\partial v}{\partial y}\right)_x \cdot \left(\frac{\partial y}{\partial v}\right)_u$

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$$x^2 = au + bv \quad \text{--- (1)}$$

$$y^2 = au - bv \quad \text{--- (2)}$$

Adding (1) and (2)

$$x^2 + y^2 = 2au$$

$$2au = x^2 + y^2$$

$$2a \left(\frac{\partial u}{\partial x}\right)_y = 2x$$

$$\left(\frac{\partial u}{\partial x}\right)_y = \frac{x}{a} \quad \text{--- (3)}$$

Using (1)

$$2x \left(\frac{\partial x}{\partial u}\right)_v = a$$

$$\left(\frac{\partial x}{\partial u}\right)_v = \frac{a}{2x} \quad \text{--- (4)}$$

Multiply (3) and (4)

$$\left(\frac{\partial u}{\partial x}\right)_y \times \left(\frac{\partial x}{\partial u}\right)_v = \frac{x}{a} \times \frac{a}{2x}$$

$$\left(\frac{\partial u}{\partial x}\right)_y \times \left(\frac{\partial x}{\partial u}\right)_v = \frac{1}{2}$$

(5)

Subtract (2) from (1)

$$x^2 - y^2 = 2bv$$

$$2bv = x^2 - y^2$$

$$2b \left(\frac{\partial v}{\partial y}\right)_x = 0 - 2y$$

$$\left(\frac{\partial v}{\partial y}\right)_x = -\frac{y}{b} \quad \text{--- (6)}$$

From (2)

$$y^2 = au - bv$$

$$2y \left(\frac{\partial y}{\partial v} \right)_u = 0 - b$$

$$\left(\frac{\partial y}{\partial v} \right)_u = -\frac{b}{2y}$$

— (7)

Multiply (6) and (7)

$$\left(\frac{\partial v}{\partial y} \right)_u \cdot \left(\frac{\partial y}{\partial v} \right)_u = \left(-\frac{1}{b} \right) \times \left(-\frac{b}{2y} \right)$$

$$\left(\frac{\partial v}{\partial y} \right)_u \cdot \left(\frac{\partial y}{\partial v} \right)_u = \frac{1}{2} \quad (8)$$

From (5) and (8)

$$\left(\frac{\partial u}{\partial v} \right)_y \cdot \left(\frac{\partial v}{\partial u} \right)_y = \frac{1}{2} = \left(\frac{\partial v}{\partial y} \right)_u \cdot \left(\frac{\partial y}{\partial v} \right)_u$$

Hence proved

DPP - 6

Topic : Partial Differentiation (Part-1)

Q.1 If $z(x+y) = x^2 + y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$

Q.2 If $f(x, y) = x^3y - xy^3$, find $\left[\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\right]_{x=1, y=2}$

Q.3 If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$ then find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$

Q.4 If $e^{\frac{z}{x^2-y^2}} = x - y$ then show that $y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} = x^2 - y^2$

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Thank You

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 7

Today's target

- Partial Differentiation (Part-2)
- PYQs
- DPP



By Gulshan sir

Q.1 If $u = x^2 \tan^{-1} \frac{y}{x} - y^2 \tan^{-1} \frac{x}{y}$ find the value of $\frac{\partial^2 u}{\partial x \partial y}$

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$$u = x^2 \tan^{-1} \frac{y}{x} - y^2 \tan^{-1} \frac{x}{y}$$

$$\frac{\partial u}{\partial y} = x^2 \times \frac{1}{1 + \left(\frac{y}{x}\right)^2} \times \frac{1}{x} - \left[y^2 \times \frac{1}{1 + \left(\frac{x}{y}\right)^2} \times \left(\frac{-x}{y^2}\right) + 2y \tan^{-1} \frac{x}{y} \right]$$

$$\frac{\partial u}{\partial y} = \frac{x}{\frac{x^2 + y^2}{x^2}} - \left[\frac{-x}{\frac{x^2 + y^2}{y^2}} + 2y \tan^{-1} \frac{x}{y} \right]$$

$$\frac{\partial u}{\partial y} = \frac{x^3}{x^2 + y^2} + \frac{xy^2}{x^2 + y^2} - 2y \tan^{-1} \frac{x}{y}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{(x^2 + y^2) 3x^2 - x^3 \times 2x}{(x^2 + y^2)^2} + \frac{(x^2 + y^2) y^2 - xy^2 \times 2x}{(x^2 + y^2)^2} - 2y \times \frac{1}{1 + \frac{x^2}{y^2}} \times \frac{1}{y}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{3x^4 + 3x^2y^2 - 2x^4}{(x^2 + y^2)^2} + \frac{x^2y^2 + y^4 - 2x^2y^2}{(x^2 + y^2)^2} - \frac{2y^2}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^4 + 3x^2y^2 + y^4 - x^2y^2 - 2y^2(x^2 + y^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^4 + 2x^2y^2 + y^4 - 2x^2y^2 - 2y^4}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^4 - y^4}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{(x^2 + y^2)(x^2 - y^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}$$

Q.2 If $u = e^{xyz}$, prove that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2 y^2 z^2)u$.

$$u = e^{xyz}$$

$$\frac{\partial u}{\partial z} = e^{xyz} \times xy$$

$$\frac{\partial^2 u}{\partial y \partial z} = e^{xyz} \times x + xy \times e^{xyz} \times xz$$

$$\frac{\partial^2 u}{\partial y \partial z} = e^{xyz} (x + x^2 yz)$$

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = e^{xyz} x(1 + 2xyz) + (x + x^2 yz) e^{xyz} \times yz$$

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = e^{xyz} \left[(1 + 2xyz) + yz(x + x^2 yz) \right]$$

$$= e^{xyz} (1 + 2xyz + xyz + x^2 y^2 z^2)$$

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = e^{xyz} (1 + 3xyz + x^2 y^2 z^2)$$

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2 y^2 z^2)u$$

Q.3 If $u = \log (x^3 + y^3 + z^3 - 3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$.

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) u$$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}\right)$$

$$u = \log(x^3 + y^3 + z^3 - 3xyz)$$

$$\frac{\partial u}{\partial x} = \frac{1}{x^3 + y^3 + z^3 - 3xyz} \times (3x^2 - 3yz)$$

$$\frac{\partial u}{\partial y} = \frac{1}{x^3 + y^3 + z^3 - 3xyz} \times (y^3 - 3xz)$$

$$\frac{\partial u}{\partial z} = \frac{1}{x^3 + y^3 + z^3 - 3xyz} \times (z^3 - 3xy)$$

Adding eqⁿ (2), (3) and (4)

$$\begin{aligned}
 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz} + \frac{3y^2 - 3xz}{x^3 + y^3 + z^3 - 3xyz} + \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz} \\
 &= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{x^3 + y^3 + z^3 - 3xyz} \\
 &= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)}
 \end{aligned}$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)} \quad (5)$$

Put eqn (5) in (1)

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \frac{3}{x+y+z}$$

$$= \frac{\partial}{\partial x} \left(\frac{3}{x+y+z} \right) + \frac{\partial}{\partial y} \left(\frac{3}{x+y+z} \right) + \frac{\partial}{\partial z} \left(\frac{3}{x+y+z} \right)$$

$$= -\frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2}$$

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$$

Q.4 If $\theta = t^n e^{-\frac{r^2}{4t}}$, Find the value of n which will make $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$

$$\theta = t^n e^{-\frac{r^2}{4t}}$$

$$\frac{\partial \theta}{\partial t} = t^n \times e^{-\frac{r^2}{4t}} \times \frac{r^2}{4t^2} + e^{-\frac{r^2}{4t}} \times n t^{n-1}$$

$$\frac{\partial \theta}{\partial t} = t^{n-1} e^{-\frac{r^2}{4t}} \left(\frac{r^2}{4t} + n \right) \text{--- ①}$$

$$\frac{\partial \theta}{\partial r} = t^n e^{-\frac{r^2}{4t}} \times \left(-\frac{2r}{4t} \right)$$

$$\frac{\partial \theta}{\partial r} = -\frac{t^{n-1}}{2} e^{-\frac{r^2}{4t}} \times r$$

$$r^2 \frac{\partial \theta}{\partial r} = -\frac{t^{n-1}}{2} r^3 e^{-\frac{r^2}{4t}}$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = -\frac{t^{n-1}}{2} \left[r^3 \times e^{-\frac{r^2}{4t}} \times \left(-\frac{2r}{2t} \right) + 3r^2 e^{-\frac{r^2}{4t}} \right]$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = -\frac{t^{n-1}}{2} r^2 e^{-\frac{r^2}{4t}} \left[-\frac{r^2}{2t} + 3 \right]$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = t^{n-1} e^{-\frac{r^2}{4t}} \left(\frac{r^2}{4t} - \frac{3}{2} \right) \text{--- ②}$$

From eqn ① and ②

$$\cancel{t^{n-1}} \cancel{e^{-\frac{x^2}{4t}}} \left(\cancel{\frac{x^2}{4t}} + n \right) = \cancel{t^{n-1}} \cancel{e^{-\frac{x^2}{4t}}} \left(\cancel{\frac{x^2}{4t}} - \frac{3}{2} \right)$$

$$\cancel{\frac{x^2}{4t}} + n = \cancel{\frac{x^2}{4t}} - \frac{3}{2}$$

$$n = -\frac{3}{2}$$

Q.5 If $u = f(r)$, where $r^2 = x^2 + y^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

$$u = f(r) \quad \text{--- (1)}$$

$$r^2 = x^2 + y^2 \quad \text{--- (2)}$$

$$2r \frac{\partial r}{\partial x} = 2x$$

$$\boxed{\frac{\partial r}{\partial x} = \frac{x}{r}}$$

$$2r \frac{\partial r}{\partial y} = 2y$$

$$\boxed{\frac{\partial r}{\partial y} = \frac{y}{r}}$$

$$u = f(r)$$

$$\frac{\partial u}{\partial r} = f'(r) \frac{\partial r}{\partial r}$$

$$\frac{\partial u}{\partial r} = \frac{r}{r} f'(r)$$

$$\frac{\partial u}{\partial r} = r \times \left[\frac{1}{r} \times f'(r) \right]$$

$$\frac{\partial^2 u}{\partial r^2} = r \times \frac{\partial}{\partial r} \left(\frac{1}{r} \times f'(r) \right) + \frac{1}{r} f'(r)$$

$$\frac{\partial^2 u}{\partial r^2} = r \left[\frac{1}{r} f''(r) \frac{\partial r}{\partial r} + f'(r) \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial r} \right] + \frac{1}{r} f'(r)$$

$$\frac{\partial^2 u}{\partial r^2} = \frac{r}{r} f''(r) \frac{r}{r} - \frac{r}{r^2} f'(r) \frac{r}{r} + \frac{1}{r} f'(r)$$

$$\frac{\partial^2 u}{\partial r^2} = \frac{r^2}{r^2} f''(r) - \frac{r^2}{r^3} f'(r) + \frac{1}{r} f'(r)$$

--- (1)

Similarly

$$\frac{\partial^2 u}{\partial y^2} = \frac{y^2}{x^2} f''(x) - \frac{y^2}{x^3} f'(x) + \frac{1}{x} f'(x) \quad \text{--- (2)}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{x^2}{x^2} f''(x) + \frac{y^2}{x^2} f''(x) - \frac{x^2}{x^3} f'(x) - \frac{y^2}{x^3} f'(x) + \frac{1}{x} f'(x) + \frac{1}{x} f'(x)$$

$$= \left(\frac{x^2 + y^2}{x^2} \right) f''(x) - \left(\frac{x^2 + y^2}{x^3} \right) f'(x) + \frac{1}{x} f'(x) + \frac{1}{x} f'(x)$$

$$= \left(\frac{x^2}{x^2} \right) f''(x) - \left(\frac{x^2}{x^3} \right) f'(x) + \frac{1}{x} f'(x) + \frac{1}{x} f'(x)$$

$$= f''(x) - \cancel{\frac{1}{x} f'(x)} + \cancel{\frac{1}{x} f'(x)} + \frac{1}{x} f'(x)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(x) + \frac{1}{x} f'(x)$$

Hence proved

DPP - 7

Topic : Partial Differentiation (Part-2)

Q.1 If $u = (x^2 + y^2 + z^2)^{-1/2}$, show that

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$$(i) \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -u \quad (ii) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Q.2 If $u = \log \sqrt{x^2 + y^2 + z^2}$, show that $(x^2 + y^2 + z^2) \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = 1$

Q.3 If $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$

Q.4 Find the value of n so that the equation $V = r^n (3 \cos^2 \theta - 1)$ satisfies the relatives

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0.$$

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Thank You

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 8

Today's target

- Euler's Theorem : For Homogeneous function of 2 variables
- PYQs
- DPP



By Gulshan sir

Homogeneous Function (For two variables)

Any function $f(x, y)$ which can be expressed in the form $x^n \phi\left(\frac{y}{x}\right)$ or $y^n \phi\left(\frac{x}{y}\right)$ is called a Homogeneous function of degree n in x and y

How to check a given function is Homogeneous or not

STEP – 1 : Put $x = tx$ and $y = ty$ in the given function $f(x, y)$

STEP – 2 : Given function $f(x, y)$ will be a Homogeneous function of degree n in x and y , if

$$f(tx, ty) = t^n f(x, y)$$

where

n = Degree of the homogeneous function

CASE – 1: If u is a Homogeneous Function of degree n in x and y , then

$$(1) \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$(2) \quad x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

CASE – 2: If u is not a Homogeneous Function but $F(u)$ Homogeneous Function of degree n in x and y , then

(Deduction from Euler's Theorem)

$$(1) \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \frac{F(u)}{F'(u)}$$

$$(2) \quad x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \phi(u)[\phi'(u) - 1]$$

Where $\phi(u) = n \frac{F(u)}{F'(u)}$

State and Prove Euler's Theorem for Homogeneous Function $f(x, y)$

Statement : if u is a Homogeneous Function of degree n in x and y , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

Proof : Homogeneous Function can be expressed as

$$u = x^n f\left(\frac{y}{x}\right)$$

$$\frac{\partial u}{\partial x} = x^n f'\left(\frac{y}{x}\right) \left(-\frac{y}{x^2}\right) + f\left(\frac{y}{x}\right) nx^{n-1}$$

$$\frac{\partial u}{\partial x} = -x^{n-2} y f'\left(\frac{y}{x}\right) + nx^{n-1} f\left(\frac{y}{x}\right)$$

Multiply both side by x

$$x \frac{\partial u}{\partial x} = -x^{n-1} y f'\left(\frac{y}{x}\right) + nx^n f\left(\frac{y}{x}\right)$$

①

$$\frac{\partial u}{\partial y} = x^n f'\left(\frac{y}{x}\right) \frac{1}{x}$$

$$\frac{\partial u}{\partial y} = x^{n-1} f'\left(\frac{y}{x}\right)$$

Multiply both side by y

$$y \frac{\partial u}{\partial y} = x^{n-1} y f'\left(\frac{y}{x}\right) \text{ ——— ②}$$

Adding ① and ②

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \cancel{-x^{n-1} y f'(\frac{y}{x})} + n x^n f(\frac{y}{x}) + \cancel{x^{n-1} y f'(\frac{y}{x})}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n x^n f(\frac{y}{x})$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n u$$

Q. 1 If $u(x, y) = (\sqrt{x} + \sqrt{y})^5$, find the value of (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

$$u(x, y) = (\sqrt{x} + \sqrt{y})^5$$

Put $x = tx$, $y = ty$

$$u(tx, ty) = (\sqrt{tx} + \sqrt{ty})^5$$

$$= (\sqrt{t} \sqrt{x} + \sqrt{t} \sqrt{y})^5$$

$$= [\sqrt{t} (\sqrt{x} + \sqrt{y})]^5$$

$$= (\sqrt{t})^5 (\sqrt{x} + \sqrt{y})^5$$

$$u(tx, ty) = t^{5/2} u(x, y)$$

Hence u is a HF of degree $n = \frac{5}{2}$

By Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu = \frac{5}{2} u$$

$$\begin{aligned} x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= n(n-1)u = \frac{5}{2} \left(\frac{5}{2} - 1 \right) u \\ &= \frac{15}{4} u \end{aligned}$$

Q.2 If $u(x, y) = f\left(\frac{y}{x}\right)$ find the value of (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$

(ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

$$u(x, y) = f\left(\frac{y}{x}\right)$$

$$u(tx, ty) = f\left(\frac{ty}{tx}\right)$$

$$u(tx, ty) = f\left(\frac{y}{x}\right)$$

$$u(tx, ty) = t^0 f\left(\frac{y}{x}\right)$$

Hence, u is a H F

of degree $n = 0$

By Euler's theorem

$$(i) x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$= 0$$

(ii)

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

$$= 0$$

Q.3 If $u = x \sin^{-1} \left(\frac{x}{y} \right) + y \sin^{-1} \left(\frac{y}{x} \right)$, evaluate $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

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$$u(x, y) = x \sin^{-1} \left(\frac{x}{y} \right) + y \sin^{-1} \left(\frac{y}{x} \right)$$

$$u(tx, ty) = tx \sin^{-1} \left(\frac{tx}{ty} \right) + ty \sin^{-1} \left(\frac{ty}{tx} \right)$$

$$u(tx, ty) = tx \sin^{-1} \left(\frac{x}{y} \right) + ty \sin^{-1} \left(\frac{y}{x} \right)$$

$$u(tx, ty) = t \left[x \sin^{-1} \left(\frac{x}{y} \right) + y \sin^{-1} \left(\frac{y}{x} \right) \right]$$

$$u(tx, ty) = t^1 u(x, y)$$

Hence, u is H.F. of degree $n = 1$

By Euler's theorem

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

$$= 1(1-1)u$$

$$= 0$$

Q.4 If $f(x, y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2 + y^2}$, prove that $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + 2f(x, y) = 0$.

$$f(x, y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2 + y^2}$$

$$f(tx, ty) = \frac{1}{(tx)^2} + \frac{1}{(tx)(ty)} + \frac{\log tx - \log ty}{(tx)^2 + (ty)^2}$$

$$f(tx, ty) = \frac{1}{t^2 x^2} + \frac{1}{t^2 xy} + \frac{\log(\frac{tx}{ty})}{t^2(x^2 + y^2)}$$

$$f(tx, ty) = \frac{1}{t^2} \left[\frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2 + y^2} \right]$$

$$f(tx, ty) = t^{-2} f(x, y)$$

Hence, f is a HF of degree $n = -2$

By Euler's theorem

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f(x, y)$$

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = -2 f(x, y)$$

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + 2 f(x, y)$$

Hence
proved

Q.5 If $u = y^2 e^{y/x} + x^2 \tan^{-1} \left(\frac{x}{y} \right)$, show that

(i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$

(ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2u$ **AKTU 2024**

$$u(x, y) = y^2 e^{y/x} + x^2 \tan^{-1} \left(\frac{x}{y} \right)$$

$$u(tx, ty) = (ty)^2 e^{\frac{tx}{ty}} + (tx)^2 \tan^{-1} \left(\frac{tx}{ty} \right)$$

$$u(tx, ty) = t^2 y^2 e^{y/x} + t^2 x^2 \tan^{-1} \left(\frac{x}{y} \right)$$

$$u(tx, ty) = t^2 \left[y^2 e^{y/x} + x^2 \tan^{-1} \left(\frac{x}{y} \right) \right]$$

$$u(tx, ty) = t^2 u(x, y)$$

Hence u is a HF of degree $n=2$

By Euler's Theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu = 2u$$

$$\begin{aligned} x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= n(n-1)u \\ &= 2(2-1)u \\ &= 2u \end{aligned}$$

Q.6 If $u = \sin^{-1} \left(\frac{x^{1/4} + y^{1/4}}{x^{1/6} + y^{1/6}} \right)$ then evaluate (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$

(ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

$$u = \sin^{-1} \left(\frac{x^{1/4} + y^{1/4}}{x^{1/6} + y^{1/6}} \right)$$

$$\sin u = \frac{x^{1/4} + y^{1/4}}{x^{1/6} + y^{1/6}}$$

$$F(u) = f(x, y)$$

Where

$$F(u) = \sin u$$

$$F'(u) = \cos u$$

$$f(x, y) = \frac{x^{1/4} + y^{1/4}}{x^{1/6} + y^{1/6}}$$

$$f(tx, ty) = \frac{(tx)^{1/4} + (ty)^{1/4}}{(tx)^{1/6} + (ty)^{1/6}}$$

$$f(tx, ty) = \frac{t^{1/4} (x^{1/4} + y^{1/4})}{t^{1/6} (x^{1/6} + y^{1/6})}$$

$$f(tx, ty) = t^{1/12} f(x, y)$$

AKTU 2017

Hence $F(u)$ is a HF of $n = \frac{1}{12}$

By Euler's Theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{n F(u)}{F'(u)}$$

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{1}{12} \times \frac{\sin u}{\cos u} \\ &= \frac{1}{12} \tan u \end{aligned}$$

Where

$$\phi(u) = \frac{1}{12} \tan u$$

$$\phi'(u) = \frac{1}{12} \sec^2 u$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \phi(u) [\phi'(u) - 1]$$

$$= \frac{\tan u}{12} \left[\frac{\sec^2 u}{12} - 1 \right]$$

$$= \frac{\tan u}{144} [\sec^2 u - 12]$$

Q.7 If $u = \cos^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$

AKTU 2019

$$u = \cos^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$$

$$\cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

$$F(u) = f(x, y)$$

Where

$$F(u) = \cos u$$

$$F'(u) = -\sin u$$

$$f(x, y) = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

$$f(tx, ty) = \frac{tx+ty}{\sqrt{tx}+\sqrt{ty}}$$

$$f(tx, ty) = \frac{t(x+y)}{\sqrt{t}(\sqrt{x}+\sqrt{y})}$$

$$f(tx, ty) = t^{1/2} f(x, y)$$

Hence, $F(u)$ is a HF of degree $n = \frac{1}{2}$

By Euler's Theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{n F(u)}{F'(u)}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \times \frac{\cos u}{-\sin u} = -\frac{1}{2} \cot u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$$

Q.8 If $u = \sec^{-1} \left(\frac{x^3 - y^3}{x + y} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$.

AKTU 2018, 2021

$$u = \sec^{-1} \left(\frac{x^3 - y^3}{x + y} \right)$$

$$\sec u = \frac{x^3 - y^3}{x + y}$$

$$F(u) = f(x, y)$$

Where

$$F(u) = \sec u$$

$$F'(u) = \sec u \tan u$$

$$f(x, y) = \frac{x^3 - y^3}{x + y}$$

$$f(tx, ty) = \frac{(tx)^3 - (ty)^3}{tx + ty}$$

$$f(tx, ty) = \frac{t^3}{t} \left(\frac{x^3 - y^3}{x + y} \right)$$

$$f(tx, ty) = t^2 f(x, y)$$

Also evaluate $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

Hence, $F(u)$ is a HF of degree
 $n = 2$

By Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{n F(u)}{F'(u)} = \frac{2 \sec u}{\sec u \tan u}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$$

Now

$$\phi(u) = 2 \cot u$$

$$\phi'(u) = -2 \operatorname{cosec}^2 u$$

$$\begin{aligned} x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= \phi(u) [\phi'(u) - 1] \\ &= 2 \cot u [-2 \operatorname{cosec}^2 u - 1] \\ &= -2 \cot u (2 \operatorname{cosec}^2 u + 1) \end{aligned}$$

Q.9 If $u = \operatorname{cosec}^{-1} \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)^{1/2}$, prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144} (13 + \tan^2 u)$

$$u = \operatorname{cosec}^{-1} \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)^{1/2}$$

$$\operatorname{cosec} u = \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)^{1/2}$$

$$F(u) = f(x, y)$$

Where

$$F(u) = \operatorname{cosec} u$$

$$F'(u) = -\operatorname{cosec} u \cot u$$

$$f(x, y) = \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)^{1/2}$$

$$f(tx, ty) = \left[\frac{(tx)^{1/2} + (ty)^{1/2}}{(tx)^{1/3} + (ty)^{1/3}} \right]^{1/2}$$

$$f(tx, ty) = \left[\frac{t^{1/2} \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)}{t^{1/3} \left(\frac{x^{1/3} + y^{1/3}}{x^{1/3} + y^{1/3}} \right)} \right]^{1/2}$$

$$f(tx, ty) = \left[t^{1/6} \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right) \right]^{1/2}$$

$$f(tx, ty) = t^{1/12} f(x, y)$$

Hence $F(u)$ is a HF of degree

$$n = \frac{1}{12}$$

$$\phi(u) = \frac{n F(u)}{F'(u)}$$

$$= \frac{1}{12} \frac{\operatorname{cosec} u}{-\operatorname{cosec} u \cot u}$$

$$= -\frac{1}{12} \tan u$$

By Euler's Theorem

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial y^2} + y^2 \frac{\partial^2 u}{\partial y^2} = \Phi(u) [\Phi'(u) - 1]$$

$$= -\frac{1}{12} \tan u \left[-\frac{1}{12} \sec^2 u - 1 \right]$$

$$= -\frac{1}{12} \tan u \left(\frac{-\sec^2 u - 12}{12} \right) = \frac{1}{12} \tan u \left(\frac{\sec^2 u + 12}{12} \right)$$

$$= \frac{\tan u}{144} (1 + \tan^2 u + 12)$$

$$= \frac{\tan u}{144} (\tan^2 u + 13) \quad \text{Hence proved}$$

Q.10 Show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u \log u$, where $\log u = \frac{x^3+y^3}{3x+4y}$

$$\log u = \frac{x^3+y^3}{3x+4y}$$

$$F(u) = f(x, y)$$

Where

$$F(u) = \log u$$

$$F'(u) = \frac{1}{u}$$

$$f(x, y) = \frac{x^3+y^3}{3x+4y}$$

$$f(tx, ty) = \frac{(tx)^3 + (ty)^3}{3(tx) + 4(ty)}$$

$$f(tx, ty) = \frac{t^3(x^3+y^3)}{t(3x+4y)}$$

$$f(tx, ty) = t^2 \left(\frac{x^3+y^3}{3x+4y} \right)$$

$$f(tx, ty) = t^2 f(x, y)$$

Hence $F(u)$ is a H.F of degree $n=2$

By Euler's Theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{n F(u)}{F'(u)} = \frac{2 \log u}{\frac{1}{u}}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u \log u$$

Hence proved

TYPE - 3 : Combination of two or more functions

Case - 1 + Case - 1 (with different degree)

Case - 2 + Case - 2

Case - 1 + Case - 2

Q.11 If $u = \frac{x^2y^2}{x^2+y^2} + \cos\left(\frac{xy}{x^2+y^2}\right)$ prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \frac{x^2y^2}{x^2+y^2}$ **AKTU 2023**

$$u = \frac{x^2y^2}{x^2+y^2} + \cos\left(\frac{xy}{x^2+y^2}\right)$$

$$\text{Let } u = v + w$$

$$v = \frac{x^2y^2}{x^2+y^2} \quad \text{--- (1)}$$

$$w = \cos\left(\frac{xy}{x^2+y^2}\right) \quad \text{--- (2)}$$

From (1)

$$v(x, y) = \frac{x^2y^2}{x^2+y^2}$$

$$v(tx, ty) = \frac{(tx)^2 (ty)^2}{(tx)^2 + (ty)^2}$$

$$= \frac{t^4 (x^2y^2)}{t^2 (x^2+y^2)}$$

$$v(tx, ty) = t^2 v(x, y)$$

Hence v is a HF of degree

$$n=2$$

By Euler's theorem

$$n^2 \frac{\partial^2 V}{\partial n^2} + 2ny \frac{\partial^2 V}{\partial n \partial y} + y^2 \frac{\partial^2 V}{\partial y^2} = n(n-1)V$$

$$= 2(2-1)V$$

$$\frac{n^2 \partial^2 V}{\partial n^2} + 2ny \frac{\partial^2 V}{\partial n \partial y} + y^2 \frac{\partial^2 V}{\partial y^2} = 2V$$

From (2) (3)

$$W(n, y) = \cos\left(\frac{ny}{n^2 + y^2}\right)$$

$$W(tn, ty) = \cos\left[\frac{tn \times ty}{(tn)^2 + (ty)^2}\right]$$

$$W(tn, ty) = \cos\left[\frac{t^2 ny}{t^2(n^2 + y^2)}\right]$$

$$W(tn, ty) = t^0 \cos\left(\frac{ny}{n^2 + y^2}\right)$$

Hence w is a HF of degree $n=0$

By Euler's theorem

$$n^2 \frac{\partial^2 w}{\partial n^2} + 2ny \frac{\partial^2 w}{\partial n \partial y} + y^2 \frac{\partial^2 w}{\partial y^2} = n(n-1)w$$

$$\frac{n^2 \partial^2 w}{\partial n^2} + 2ny \frac{\partial^2 w}{\partial n \partial y} + y^2 \frac{\partial^2 w}{\partial y^2} = 0 \quad (4)$$

Adding ③ and ④

$$x^2 \frac{\partial^2}{\partial x^2}(v+w) + 2xy \frac{\partial^2}{\partial x \partial y}(v+w) + y^2 \frac{\partial^2}{\partial y^2}(v+w) = 2v + 0$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \frac{x^2 y^2}{x^2 + y^2}$$

Hence prove

Q.12 $u = \frac{x^4 - y^4}{x^2 y^2} + \cos^{-1} \frac{x+y}{\sqrt{x} + \sqrt{y}}$ then find the value of (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

$$u = \frac{x^4 - y^4}{x^2 y^2} + \cos^{-1} \frac{x+y}{\sqrt{x} + \sqrt{y}}$$

Let $u = v + w$

$$v = \frac{x^4 - y^4}{x^2 y^2} \quad \text{--- (1)}$$

$$w = \cos^{-1} \frac{x+y}{\sqrt{x} + \sqrt{y}} \quad \text{--- (2)}$$

From (1)

$$v(x, y) = \frac{x^4 - y^4}{x^2 y^2}$$

$$v(tx, ty) = \frac{(tx)^4 - (ty)^4}{(tx)^2 (ty)^2}$$

$$v(tx, ty) = \frac{t^4}{t^4} \left(\frac{x^4 - y^4}{x^2 y^2} \right)$$

$$v(tx, ty) = t^0 v(x, y)$$

Hence v is a H.F of degree $n = 0$

By Euler's Theorem

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = nv$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 0$$

$$\text{--- (3)}$$

$$x^2 \frac{\partial^2 v}{\partial x^2} + 2xy \frac{\partial^2 v}{\partial x \partial y} + y^2 \frac{\partial^2 v}{\partial y^2} = n(n-1)v$$

$$x^2 \frac{\partial^2 v}{\partial x^2} + 2xy \frac{\partial^2 v}{\partial x \partial y} + y^2 \frac{\partial^2 v}{\partial y^2} = 0$$

From (2)

$$w = \cos^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$$

$$\cos w = \frac{x+y}{\sqrt{x} + \sqrt{y}}$$

$$F(w) = f(x, y)$$

Where

(4)

$$F(w) = \cos w$$

$$F'(w) = -\sin w$$

$$f(x, y) = \frac{x+y}{\sqrt{x} + \sqrt{y}}$$

$$f(tx, ty) = \frac{tx + ty}{\sqrt{tx} + \sqrt{ty}} = \frac{t(x+y)}{\sqrt{t}(\sqrt{x} + \sqrt{y})}$$

$$f(tx, ty) = t^{1/2} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$$

$$f(tx, ty) = t^{1/2} f(x, y)$$

Hence, $\cos w$ is a H.F of degree $n = 1/2$

By Euler's Theorem

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = \frac{n F(w)}{F'(w)} = \frac{1}{2} \times \frac{\cos w}{-\sin w}$$

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = -\frac{1}{2} \cot w \quad \text{--- (5)}$$

$$\therefore \phi(w) = -\frac{1}{2} \cot w$$

$$x^2 \frac{\partial^2 w}{\partial x^2} + 2xy \frac{\partial^2 w}{\partial x \partial y} + y^2 \frac{\partial^2 w}{\partial y^2} = \phi(w) [\phi'(w) - 1]$$

$$x^2 \frac{\partial^2 w}{\partial x^2} + 2xy \frac{\partial^2 w}{\partial x \partial y} + y^2 \frac{\partial^2 w}{\partial y^2} = -\frac{1}{2} \cot w \left[\frac{1}{2} \operatorname{cosec}^2 w - 1 \right] \quad \text{--- (6)}$$

Adding (3) and (5)

$$x \frac{\partial}{\partial x} (v+w) + y \frac{\partial}{\partial y} (v+w) = -\frac{1}{2} \cot w + 0$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \cot w \quad \checkmark$$

Adding (4) and (6)

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{1}{2} \cot w \left[\frac{1}{2} \operatorname{cosec}^2 w - 1 \right]$$

Where

$$w = \cos^{-1} \frac{x+y}{\sqrt{x} + \sqrt{y}}$$

TYPE - 4 : Verification of Euler's theorem

Q.13 Verify Euler's theorem for $z = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$

AKTU 2016, 2018

$$Z(x, y) = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$$

$$Z(tx, ty) = \frac{(tx)^{1/3} + (ty)^{1/3}}{(tx)^{1/2} + (ty)^{1/2}}$$

$$Z(tx, ty) = \frac{t^{1/3}(x^{1/3} + y^{1/3})}{t^{1/2}(x^{1/2} + y^{1/2})}$$

$$Z(tx, ty) = t^{-1/6} \left(\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}} \right)$$

$$Z(tx, ty) = t^{-1/6} Z(x, y)$$

Hence, Z is a HF of degree $n = -\frac{1}{6}$

By Euler's theorem

$$x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} = nZ$$

$$x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} = -\frac{1}{6} Z$$

①

$$Z = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$$

Taking log both side

$$\log Z = \log \left(\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}} \right)$$

$$\log Z = \log(x^{1/3} + y^{1/3}) - \log(x^{1/2} + y^{1/2}) \quad \text{--- (2)}$$

$$\frac{1}{Z} \frac{\partial Z}{\partial x} = \frac{\frac{1}{3} x^{-2/3}}{x^{1/3} + y^{1/3}} - \frac{\frac{1}{2} x^{-1/2}}{x^{1/2} + y^{1/2}}$$

$$\frac{x}{Z} \frac{\partial Z}{\partial x} = \frac{\frac{1}{3} x^{1/3}}{x^{1/3} + y^{1/3}} - \frac{\frac{1}{2} x^{1/2}}{x^{1/2} + y^{1/2}} \quad \text{--- (3)}$$

$$\frac{1}{Z} \frac{\partial Z}{\partial y} = \frac{\frac{1}{3} y^{-2/3}}{x^{1/3} + y^{1/3}} - \frac{\frac{1}{2} y^{-1/2}}{x^{1/2} + y^{1/2}}$$

$$\frac{y}{Z} \frac{\partial Z}{\partial y} = \frac{\frac{1}{3} y^{1/3}}{x^{1/3} + y^{1/3}} - \frac{\frac{1}{2} y^{1/2}}{x^{1/2} + y^{1/2}} \quad \text{--- (4)}$$

Adding (3) and (4)

$$\frac{x}{Z} \frac{\partial Z}{\partial x} + \frac{y}{Z} \frac{\partial Z}{\partial y} = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$$

$$\boxed{x \frac{\partial Z}{\partial x} + y \frac{\partial Z}{\partial y} = -\frac{1}{6} Z} \quad \text{--- (5)}$$

Hence, Euler's theorem is verified

UNIT -2 : DPP - 8

Topic : Euler's Theorem for Homogeneous Function of 2 variables

TYPE - 1 : Apply Case - 1

Q.1 If $u = (x^{1/4} + y^{1/4}) \left(x^{1/5} + y^{1/5} \right)$, apply Euler's theorem to find the value of

(i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$

(ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

Q.2 If $u(x, y) = x^2 y^5 f\left(\frac{y}{x}\right)$ find the value of (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

Q.3 If $f(x, y) = \sqrt{x^2 - y^2} \sin^{-1} \frac{y}{x}$, prove that $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = f(x, y)$

Q.4 If $f(x, y) = \sqrt{y^2 - x^2} \sin^{-1} \frac{x}{y} + \frac{x^2 - y^2}{\sqrt{x^2 - y^2}}$, prove that $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} - f(x, y) = 0$.

Q.5 If $u = x^2 \tan^{-1} \left(\frac{y}{x} \right) - y^2 \tan^{-1} \left(\frac{x}{y} \right)$, evaluate $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

Q.6 If $u = \sin^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$ prove that

$$(i) \ x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u \quad (ii) \ x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\sin u \cos 2u}{4 \cos^3 u}$$

Q.7 If $u = \sin^{-1} \left(\frac{x^{1/3}+y^{1/3}}{x^{1/2}-y^{1/2}} \right)^{1/2}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{12} \tan u$

Q.8 If $u = \tan^{-1} \frac{x^3+y^3}{x-y}$, prove that

$$(i) \ x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u \quad (ii) \ x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \sin 4u - \sin 2u = 2 \cos 3u \sin u.$$

Q.9 If $u = \log_e \left(\frac{x^4+y^4}{x+y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$

Q.10 Prove that $xu_x + yu_y = \frac{5}{2} \tan u$ if $u = \sin^{-1} \left(\frac{x^3+y^3}{\sqrt{x}+\sqrt{y}} \right)$

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Case – 1 + Case – 1 (with different degree)

Case – 2 + Case – 2

Case – 1 + Case – 2

Q.11 If $z = x^4 y^2 \sin^{-1} \left(\frac{x}{y} \right) + \log x - \log y$, show that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 6x^4 y^2 \sin^{-1} \left(\frac{x}{y} \right)$.

Q.12 If $u = x\phi \left(\frac{y}{x} \right) + \Psi \left(\frac{y}{x} \right)$, show that

$$(i) \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = x\phi \left(\frac{y}{x} \right)$$

$$(ii) \quad x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$$

Q.13 If $u = x^n f_1 \left(\frac{y}{x} \right) + y^{-n} f_2 \left(\frac{x}{y} \right)$, prove that $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n^2 z$

TYPE – 4 : Verification of Euler's theorem

Q.14 Verify Euler's theorem for $u = \frac{x(x^4 - y^4)}{x^4 + y^4}$

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Q.15 Verify Euler's theorem for $u = \log \left(\frac{x^4 - y^4}{x + y} \right)$

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Thank You

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 9

Today's target

- Euler's Theorem : For Homogeneous function of 3 variables
- PYQs
- DPP



By Gulshan sir

Homogeneous Function (For three variables)

Any function $f(x, y, z)$ which can be expressed in the form $x^n \phi\left(\frac{y}{x}, \frac{z}{x}\right)$ or $y^n \phi\left(\frac{x}{y}, \frac{z}{y}\right)$ is called a

Homogeneous Function of degree n in x, y and z

How to check a given function is Homogeneous or not

STEP – 1 : Put $x = tx$, $y = ty$ and $z = tz$ in the given function $f(x, y, z)$

STEP – 2 : Given function $f(x, y, z)$ will be a Homogeneous function of degree n in x, y and z , if

$$f(tx, ty, tz) = t^n f(x, y, z)$$

Where

n = Degree of the Homogeneous function

NOTE : Euler's theorem can be extended to any number of variables

CASE – 1: If u is a Homogeneous Function of degree n in x , y and z , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$$

CASE – 2: If u is not a Homogeneous Function but $F(u)$ is a Homogeneous Function of degree n in x , y and z , then

(Deduction from Euler's Theorem)

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = n \frac{F(u)}{F'(u)}$$

NOTE : Euler's theorem can be extended to any number of variables

Q.1 If $u = \frac{x^4+y^4+z^4}{x+y+z}$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3u$

$$u(x, y, z) = \frac{x^4 + y^4 + z^4}{x + y + z}$$

Put $x = tx, y = ty, z = tz$

$$u(tx, ty, tz) = \frac{(tx)^4 + (ty)^4 + (tz)^4}{(tx) + (ty) + (tz)}$$

$$= \frac{t^4 (x^4 + y^4 + z^4)}{t(x + y + z)}$$

$$u(tx, ty, tz) = t^3 u(x, y, z)$$

Hence u is a HF of degree $n = 3$

By Euler's Theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3u$$

Q.2 Show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 2 \tan u$, where $u = \sin^{-1} \left(\frac{x^3 + y^3 + z^3}{ax + by + cz} \right)$

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$$u = \sin^{-1} \left(\frac{x^3 + y^3 + z^3}{ax + by + cz} \right)$$

$$f(x, y, z) = \frac{x^3 + y^3 + z^3}{ax + by + cz}$$

$$\sin u = \frac{x^3 + y^3 + z^3}{ax + by + cz}$$

$$f(tx, ty, tz) = \frac{(tx)^3 + (ty)^3 + (tz)^3}{a(tx) + b(ty) + c(tz)}$$

$$F(u) = f(x, y, z)$$

Where

$$F(u) = \sin u$$

$$F'(u) = \cos u$$

$$f(tx, ty, tz) = \frac{t^3}{t} \left(\frac{x^3 + y^3 + z^3}{x + y + z} \right)$$

$$f(tx, ty, tz) = t^2 f(x, y, z)$$

Hence, $F(u)$ is a HF of degree $n=2$

By Euler's Theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = n \frac{F(u)}{F'(u)}$$

$$= 2 \frac{\sin u}{\cos u}$$

$$= 2 \tan u$$

Hence proved

Case - 1 + Case - 1 (with different degree)

Case - 2 + Case - 2

Case - 1 + Case - 2

Q.3 If $u = \frac{x^2 y^2 z^2}{x^2 + y^2 + z^2} + \cos \frac{xy + yz}{x^2 + y^2 + z^2}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{4x^2 y^2 z^2}{x^2 + y^2 + z^2}$

$$u = \frac{x^2 y^2 z^2}{x^2 + y^2 + z^2} + \cos \frac{xy + yz}{x^2 + y^2 + z^2}$$

Let $u = v + w$

$$v = \frac{x^2 y^2 z^2}{x^2 + y^2 + z^2} \quad \text{--- (1)}$$

$$w = \cos \left(\frac{xy + yz}{x^2 + y^2 + z^2} \right) \quad \text{--- (2)}$$

From (1)

$$v(x, y, z) = \frac{x^2 y^2 z^2}{x^2 + y^2 + z^2}$$

$$v(tx, ty, tz) = \frac{(tx)^2 (ty)^2 (tz)^2}{(tx)^2 + (ty)^2 + (tz)^2}$$

$$v(tx, ty, tz) = \frac{t^6}{t^2} \left(\frac{x^2 y^2 z^2}{x^2 + y^2 + z^2} \right)$$

$$v(t_n, t_y, t_z) = t^4 v(n, y, z)$$

Hence v is HF of degree
 $n=4$

By Euler's Theorem

$$n \frac{\partial v}{\partial n} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = nv$$

$$n \frac{\partial v}{\partial n} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = 4v$$

③

From ②

$$w(n, y, z) = \cos\left(\frac{ny + yz}{n^2 + y^2 + z^2}\right)$$

$$w(t_n, t_y, t_z) = \cos\left[\frac{t_n t_y + t_y t_z}{(t_n)^2 + (t_y)^2 + (t_z)^2}\right]$$

$$= \cos\left[\frac{t^4}{t^4} \left(\frac{ny + yz}{n^2 + y^2 + z^2}\right)\right]$$

$$= t^0 \cos\left(\frac{ny + yz}{n^2 + y^2 + z^2}\right)$$

$$= t^0 w(n, y, z)$$

Hence w is a Homogeneous function of degree $n=0$

By Euler's Theorem

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} + z \frac{\partial w}{\partial z} = nw = 0 \quad \text{--- (4)}$$

Adding (3) and (4)

$$x \frac{\partial}{\partial x}(v+w) + y \frac{\partial}{\partial y}(v+w) + z \frac{\partial}{\partial z}(v+w) = 4v + 0$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{4x^2y^2z^2}{x^2+y^2+z^2}$$

Hence proved

TYPE - 4 : Verification of Euler's theorem

Q.4 Verify Euler's theorem for the following function $u = x^3 + y^3 + z^3 + 3xyz$

$$u(x, y, z) = x^3 + y^3 + z^3 + 3xyz$$

$$u(tx, ty, tz) = (tx)^3 + (ty)^3 + (tz)^3 + 3(tx)(ty)(tz)$$

$$u(tx, ty, tz) = t^3(x^3 + y^3 + z^3 + 3xyz)$$

$$u(tx, ty, tz) = t^3 u(x, y, z)$$

Hence u is a HF of degree

$$n=3$$

By Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3u$$

①

$$u = x^3 + y^3 + z^3 + 3xyz$$

$$\frac{\partial u}{\partial x} = 3x^2 + 3yz$$

$$x \frac{\partial u}{\partial x} = 3x^3 + 3xyz \quad \text{--- (2)}$$

$$\frac{\partial u}{\partial y} = 3y^2 + 3xz$$

$$y \frac{\partial u}{\partial y} = 3y^3 + 3xyz \quad \text{--- (3)}$$

$$\frac{\partial u}{\partial z} = 3z^2 + 3xy$$

$$z \frac{\partial u}{\partial z} = 3z^3 + 3xyz \quad \text{--- (4)}$$

Adding (2), (3) and (4)

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} &= 3x^3 + 3xyz + 3y^3 + 3xyz + 3z^3 + 3xyz \\ &= 3(x^3 + y^3 + z^3 + 3xyz) \end{aligned}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3u \quad \text{--- (5)}$$

From (1) and (5)

Euler's theorem is verified

UNIT -2 : DPP - 9

Topic : Euler's Theorem for Homogeneous Function of 3 variables

Q.1 If $u = \cos \left(\frac{xy+yz+zx}{x^2+y^2+z^2} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$

Q.2 If $u = \log \left(\frac{x^5+y^5+z^5}{x^2+y^2+z^2} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3$

Q.3 Show that $xu_x + yu_y + zu_z = -2 \cot u$, where $u = \cos^{-1} \left(\frac{x^3+y^3+z^3}{ax+by+cz} \right)$

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Q.4 If $u = \sin^{-1} \left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} + 3 \tan u = 0$

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Thank You

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 10

Today's target

- Total Differentiation (Part-1)
- PYQs
- DPP



By Gulshan sir

Total Differentiation : Differentiation of Composite functions

CASE – 1 : If u is a composite function of single variable t , such that

$$u = f(x, y)$$

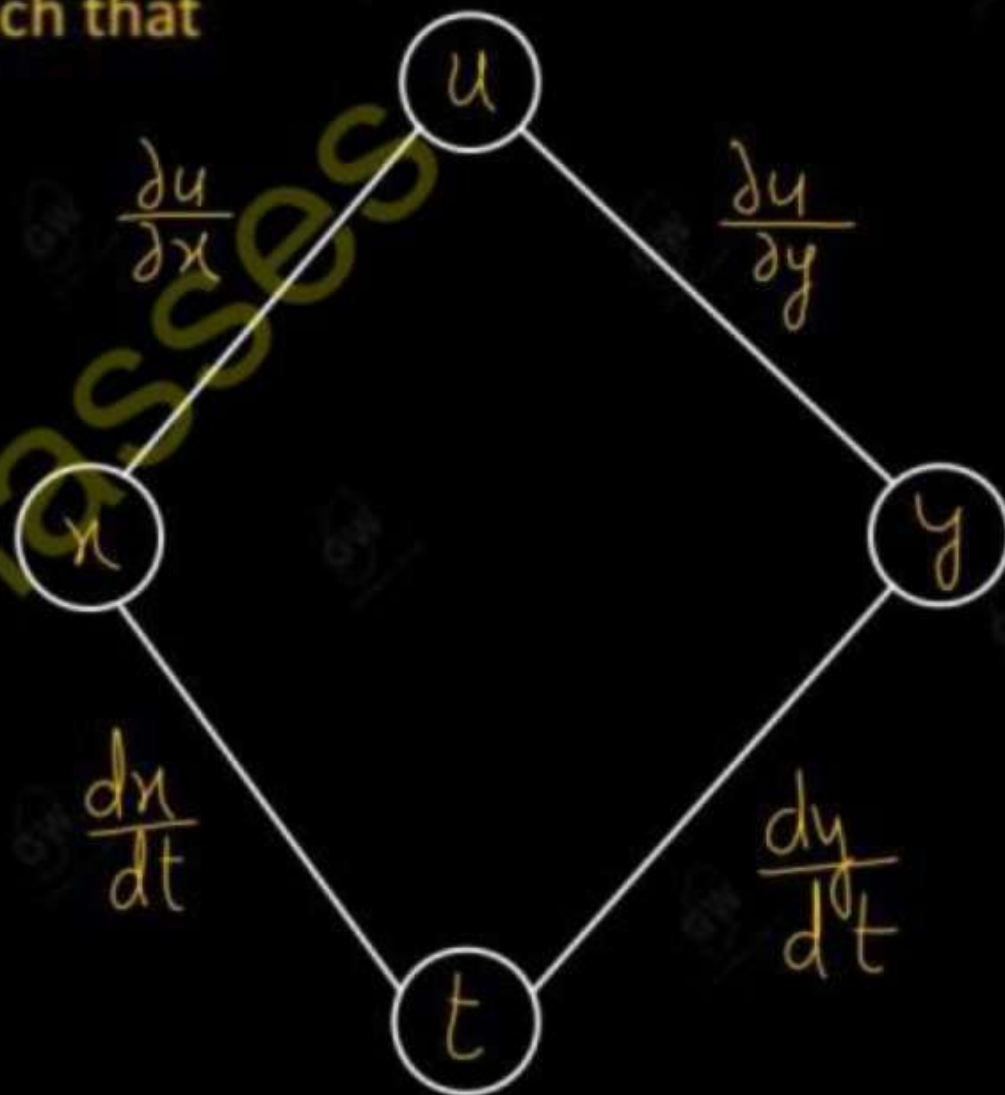
$$x = \phi_1(t)$$

$$y = \phi_2(t)$$

$$u = f(t)$$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$



Q.1 If $u = x^3 + y^3$, where $x = a \cos t$, $y = a \sin t$ find $\frac{du}{dt}$ and verify the result.

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$$u = x^3 + y^3$$

$$x = a \cos t$$

$$y = a \sin t$$

u is a composite function of t

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$\frac{du}{dt} = 3x^2 \times (-a \sin t) + 3y^2 a \cos t$$

$$\frac{du}{dt} = -3(a \cos t)^2 (a \sin t) + 3(a \sin t)^2 a \cos t$$

$$\frac{du}{dt} = -3a^3 \cos^2 t \sin t + 3a^3 \sin^2 t \cos t$$

①

Verification

$$u = x^3 + y^3$$

$$u = a^3 \cos^3 t + a^3 \sin^3 t$$

$$\frac{du}{dt} = a^3 \times 3 \cos^2 t (-\sin t) + a^3 \times 3 \sin^2 t \cos t$$

$$\frac{du}{dt} = -3a^3 \cos^2 t \sin t + 3a^3 \sin^2 t \cos t$$

②

From ① and ② Result is verified

CASE – 2 : If u is a composite function of single variable t , such that

$$u = f(x, y, z)$$

$$u = f(t)$$

$$x = \phi_1(t)$$

$$y = \phi_2(t)$$

$$z = \phi_3(t)$$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$



Q. 2 Find $\frac{du}{dt}$ as a total derivative and verify the result by direct substitution if

$$u = x^2 + y^2 + z^2$$

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and $x = e^{2t}$, $y = e^{2t} \cos 3t$, $z = e^{2t} \sin 3t$

$$u = x^2 + y^2 + z^2$$

$$x = e^{2t}$$

$$y = e^{2t} \cos 3t$$

$$z = e^{2t} \sin 3t$$

u is a composite function of t

$$\frac{du}{dt} = 2x \times 2e^{2t} + 2y [e^{2t}(-3\sin 3t) + \cos 3t \times 2e^{2t}] + 2z [e^{2t}(3\cos 3t) + \sin 3t \times 2e^{2t}]$$

$$\frac{du}{dt} = 2e^{2t} \times 2e^{2t} + 2e^{2t} \cos 3t [-3e^{2t} \sin 3t + 2e^{2t} \cos 3t] + 2e^{2t} \sin 3t [3e^{2t} \cos 3t + 2e^{2t} \sin 3t]$$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

$$\frac{du}{dt} = 4e^{4t} - 6e^{4t} \cos 3t \sin 3t + 4e^{4t} \cos^2 3t + 6e^{4t} \sin 3t \cos 3t + 4e^{4t} \sin^2 3t$$

$$\frac{du}{dt} = 4e^{4t} + 4e^{4t}(\cos^2 3t + \sin^2 3t) = 4e^{4t} + 4e^{4t}$$

$$\boxed{\frac{du}{dt} = 8e^{4t}} \quad \text{--- (1)}$$

Verification

$$u = x^2 + y^2 + z^2$$

$$u = (e^{2t})^2 + (e^{2t} \cos 3t)^2 + (e^{2t} \sin 3t)^2$$

$$u = e^{4t} + e^{4t} \cos^2 3t + e^{4t} \sin^2 3t$$

$$u = e^{4t} + e^{4t}(\cos^2 3t + \sin^2 3t)$$

$$u = e^{4t} + e^{4t}$$

$$u = 2e^{4t}$$

$$\frac{du}{dt} = 2 \times 4e^{4t}$$

$$\boxed{\frac{du}{dt} = 8e^{4t}} \quad \text{--- (2)}$$

From (1) and (2) Result is verified

UNIT -2 : DPP - 10

Topic : Total Differentiation (Part-1)

Q.1 If $z = u^3 + v^3$, where $u = at^2, v = 2at$ find $\frac{dz}{dt}$ and verify the result.

Q.2 If $u = \sin^{-1}(x - y), x = 3t, y = 4t^2$ show that $\frac{du}{dt} = \frac{3}{\sqrt{1-t^2}}$

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Thank You

ENGINEERING MATHEMATICS - I

UNIT -2 : Differential calculus - I

Lecture - 11

Today's target

- Total Differentiation (Part - 2)
- PYQs
- DPP



By Gulshan sir

Total Differentiation : Differentiation of Composite functions

CASE – 3 : If z is a composite function of u and v such that

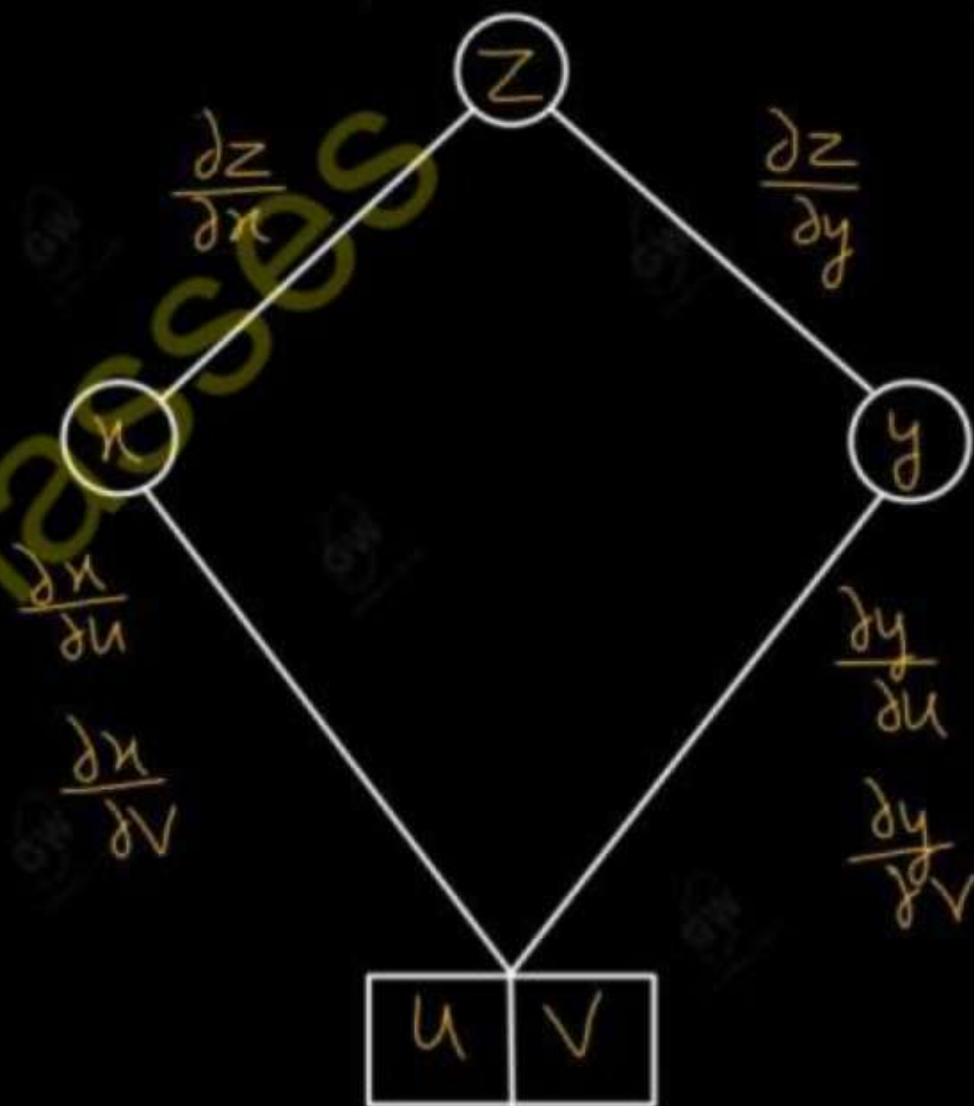
$$z = f(x, y)$$

$$x = \phi_1(u, v)$$

$$y = \phi_2(u, v)$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$



Q.1 If z is a function of x and y , where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$ show that

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$$

AKTU-2024

$$z = f(x, y)$$

$$x = e^u + e^{-v}$$

$$y = e^{-u} - e^v$$

z is a composite function of u and v

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} e^u - \frac{\partial z}{\partial y} e^{-u} \quad \text{--- (1)}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial v} = -\frac{\partial z}{\partial x} e^{-v} - \frac{\partial z}{\partial y} e^v \quad \text{--- (2)}$$

Subtract (2) from (1)

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = e^u \frac{\partial z}{\partial x} - e^{-u} \frac{\partial z}{\partial y} + e^{-v} \frac{\partial z}{\partial x} + e^v \frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = (e^u + e^{-v}) \frac{\partial z}{\partial x} - (e^{-u} - e^v) \frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$$

Hence proved

Q.2 If $w = f(x, y)$ where $x = e^u \cos v$, $y = e^u \sin v$ show that $y \frac{\partial w}{\partial u} + x \frac{\partial w}{\partial v} = e^{2u} \frac{\partial w}{\partial y}$

$$w = f(x, y)$$

$$x = e^u \cos v$$

$$y = e^u \sin v$$

w is a composite function of u and v

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial w}{\partial u} = e^u \cos v \frac{\partial w}{\partial x} + e^u \sin v \frac{\partial w}{\partial y}$$

$$y \frac{\partial w}{\partial u} = y e^u \cos v \frac{\partial w}{\partial x} + y e^u \sin v \frac{\partial w}{\partial y}$$

$$y \frac{\partial w}{\partial u} = e^{2u} \sin v \cos v \frac{\partial w}{\partial x} + e^{2u} \sin^2 v \frac{\partial w}{\partial y} \quad \text{--- (1)}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v}$$

$$\frac{\partial w}{\partial v} = -e^u \sin v \frac{\partial w}{\partial x} + e^u \cos v \frac{\partial w}{\partial y}$$

$$x \frac{\partial w}{\partial v} = -x e^u \sin v \frac{\partial w}{\partial x} + x e^u \cos v \frac{\partial w}{\partial y}$$

$$x \frac{\partial w}{\partial v} = -e^{2u} \sin v \cos v \frac{\partial w}{\partial x} + e^{2u} \cos^2 v \frac{\partial w}{\partial y} \quad \text{--- (2)}$$

Adding (1) and (2)

$$y \frac{\partial w}{\partial u} + x \frac{\partial w}{\partial v} = \cancel{e^{2u} \sin v \cos v \frac{\partial w}{\partial u}} + e^{2u} \sin^2 v \frac{\partial w}{\partial y} - \cancel{e^{2u} \sin v \cos v \frac{\partial w}{\partial u}} + e^{2u} \cos^2 v \frac{\partial w}{\partial y}$$

$$y \frac{\partial w}{\partial u} + x \frac{\partial w}{\partial v} = e^{2u} \frac{\partial w}{\partial y} (\sin^2 v + \cos^2 v)$$

$$y \frac{\partial w}{\partial u} + x \frac{\partial w}{\partial v} = e^{2u} \frac{\partial w}{\partial y}$$

Hence proved

Q.3 If $w = f(x, y)$ where $x = r \cos \theta$, $y = r \sin \theta$, show that $\left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2$

$$w = f(x, y)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

w is a composite function of r and θ

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}$$

$$\frac{\partial w}{\partial x} = \cos \theta \frac{\partial w}{\partial r} + \sin \theta \frac{\partial w}{\partial \theta} \quad \text{--- (1)}$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta}$$

$$\frac{\partial w}{\partial \theta} = (-r \sin \theta) \frac{\partial w}{\partial x} + r \cos \theta \frac{\partial w}{\partial y}$$

$$\frac{1}{r} \frac{\partial w}{\partial \theta} = -\sin \theta \frac{\partial w}{\partial x} + \cos \theta \frac{\partial w}{\partial y} \quad \text{--- (2)}$$

Squaring and adding (1) and (2)

$$\left(\frac{\partial w}{\partial x}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\cos \theta \frac{\partial w}{\partial r} + \sin \theta \frac{\partial w}{\partial \theta}\right)^2 + \left(-\sin \theta \frac{\partial w}{\partial r} + \cos \theta \frac{\partial w}{\partial \theta}\right)^2$$

$$\begin{aligned} \left(\frac{\partial w}{\partial x}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 &= \cos^2 \theta \left(\frac{\partial w}{\partial n}\right)^2 + \sin^2 \theta \left(\frac{\partial w}{\partial y}\right)^2 + \cancel{2 \sin \theta \cos \theta \frac{\partial w}{\partial n} \frac{\partial w}{\partial y}} + \sin^2 \theta \left(\frac{\partial w}{\partial n}\right)^2 \\ &\quad + \cos^2 \theta \left(\frac{\partial w}{\partial y}\right)^2 - \cancel{2 \sin \theta \cos \theta \frac{\partial w}{\partial n} \frac{\partial w}{\partial y}} \\ &= (\cos^2 \theta + \sin^2 \theta) \left(\frac{\partial w}{\partial n}\right)^2 + (\sin^2 \theta + \cos^2 \theta) \left(\frac{\partial w}{\partial y}\right)^2 \end{aligned}$$

$$\left(\frac{\partial w}{\partial x}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\frac{\partial w}{\partial n}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2$$

$$\boxed{\left(\frac{\partial w}{\partial x}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\frac{\partial f}{\partial n}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Hence proved

$$\left\{ \because \frac{\partial w}{\partial n} = \frac{\partial f}{\partial n} \right\}$$

CASE — 4 : If u is a composite function of x, y and z such that

$$u = f(r, s, t)$$

$$r = \phi_1(x, y, z)$$

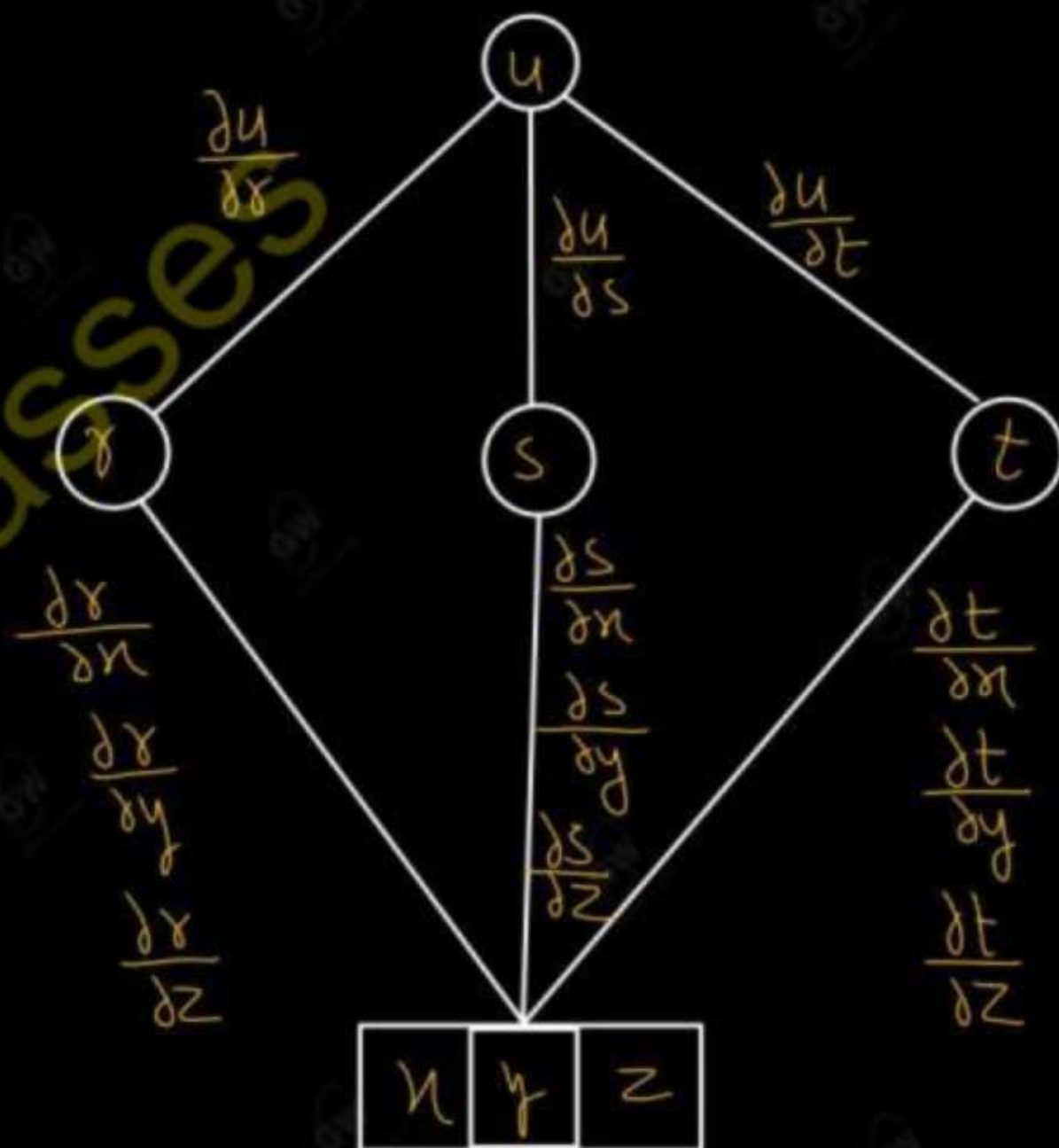
$$s = \phi_2(x, y, z)$$

$$t = \phi_3(x, y, z)$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z}$$



$$u = f(y - z, z - x, x - y)$$

$$\text{Let } u = f(r, s, t)$$

$$r = y - z$$

$$s = z - x$$

$$t = x - y$$

u is a composite function of x, y, z

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial x} = 0 - \frac{\partial u}{\partial s} + \frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} + 0 - \frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \quad \text{--- (2)}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z}$$

$$\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} + 0$$

$$\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \quad \text{--- (3)}$$

Adding ①, ② and ③

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = -\cancel{\frac{\partial u}{\partial s}} + \cancel{\frac{\partial u}{\partial t}} + \cancel{\frac{\partial u}{\partial r}} - \cancel{\frac{\partial u}{\partial t}} - \cancel{\frac{\partial u}{\partial r}} + \cancel{\frac{\partial u}{\partial s}}$$

$$\boxed{\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0}$$

Hence proved

Q.5 If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$

AKTU-2015, 2020

$$u = f(2x - 3y, 3y - 4z, 4z - 2x)$$

$$\text{Let } u = f(r, s, t)$$

$$r = 2x - 3y$$

$$s = 3y - 4z$$

$$t = 4z - 2x$$

u is a composite function of x, y, z

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial r} + 0 - 2 \frac{\partial u}{\partial t}$$

$$\frac{1}{2} \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y}$$

$$\frac{\partial u}{\partial y} = -3 \frac{\partial u}{\partial r} + 3 \frac{\partial u}{\partial s} + 0$$

$$\frac{1}{3} \frac{\partial u}{\partial y} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \quad \text{--- (2)}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z}$$

$$\frac{\partial u}{\partial z} = 0 - 4 \frac{\partial u}{\partial s} + 4 \frac{\partial u}{\partial t}$$

$$\frac{1}{4} \frac{\partial u}{\partial z} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \quad \text{--- (3)}$$

Adding (1), (2) and (3)

$$\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = \cancel{\frac{\partial u}{\partial x}} - \cancel{\frac{\partial u}{\partial t}} - \cancel{\frac{\partial u}{\partial x}} + \cancel{\frac{\partial u}{\partial s}} - \cancel{\frac{\partial u}{\partial s}} + \cancel{\frac{\partial u}{\partial t}}$$

$$\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$$

Hence proved

Q.6 If $u = f(r, s, t)$ and $r = \frac{x}{y}, s = \frac{y}{z}, t = \frac{z}{x}$, prove that $x \frac{du}{dx} + y \frac{du}{dy} + z \frac{du}{dz} = 0$

(AKTU-2018)

$$u = f(r, s, t)$$

$$r = \frac{x}{y}$$

$$s = \frac{y}{z}$$

$$t = \frac{z}{x}$$

u is a composite function of x, y, z

$$\frac{du}{dx} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x}$$

$$\frac{du}{dx} = \frac{\partial u}{\partial r} \frac{1}{y} + 0 - \frac{z}{x^2} \frac{\partial u}{\partial t}$$

$$x \frac{du}{dx} = \frac{x}{y} \frac{\partial u}{\partial r} - \frac{z}{x} \frac{\partial u}{\partial t} \quad \text{--- (1)}$$

$$\frac{du}{dy} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y}$$

$$\frac{du}{dy} = -\frac{x}{y^2} \frac{\partial u}{\partial r} + \frac{1}{z} \frac{\partial u}{\partial s} + 0$$

$$y \frac{du}{dy} = -\frac{x}{y} \frac{\partial u}{\partial r} + \frac{y}{z} \frac{\partial u}{\partial s} \quad \text{--- (2)}$$

$$\frac{z}{x} \frac{du}{dz} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z}$$

$$\frac{\partial u}{\partial z} = 0 - \frac{y}{z^2} \frac{\partial u}{\partial s} + \frac{1}{x} \frac{\partial u}{\partial t}$$

$$z \frac{\partial u}{\partial z} = -\frac{y}{z} \frac{\partial u}{\partial s} + \frac{z}{x} \frac{\partial u}{\partial t} \quad \text{--- (3)}$$

Adding ①, ② and ③

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{x}{y} \frac{\partial u}{\partial x} - \frac{z}{x} \frac{\partial u}{\partial t} - \frac{x}{y} \frac{\partial u}{\partial x} + \frac{y}{z} \frac{\partial u}{\partial s} - \frac{y}{z} \frac{\partial u}{\partial s} + \frac{z}{x} \frac{\partial u}{\partial t}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

Hence proved

UNIT -2 : DPP - 11

Topic : Total Differentiation (Part - 2)

Q.1 If $u = f(r, s)$ and $r = x + y, s = x - y$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2 \frac{\partial u}{\partial r}$

Q.2 If $z = z(u, v)$ and $u = x^2 - 2xy - y^2, v = y$ show that $(x + y) \frac{\partial z}{\partial x} + (x - y) \frac{\partial z}{\partial y} = 0$

Q.3 If $z = f(x, y)$, where $x = e^u \cos v, y = e^u \sin v$, prove that $\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 = e^{-2u} \left[\left(\frac{\partial f}{\partial u}\right)^2 + \left(\frac{\partial f}{\partial v}\right)^2\right]$

Q.4 If $u = f(e^{y-z} + e^{z-x} + e^{x-y})$ prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Q.5 If $w = \sqrt{x^3 + y^3 + z^3}$ and $x = u \cos v, y = u \sin v, z = uv$, then prove that $u \frac{\partial w}{\partial u} - v \frac{\partial w}{\partial v} = \frac{u}{\sqrt{1+v^2}}$.

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Thank You